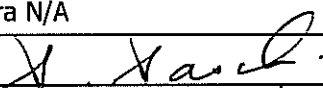
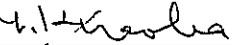
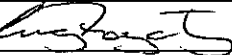
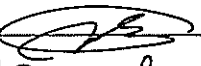


DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

Contract No.: DM001			CDRL No.:	
Project Title: DUBAI METRO			CDRL Title:	
Document Title: Jebel Ali Auxiliary Depot Structural Design Calculation Building 11 - DCP 3 Submission				
Revision History				
A3	15-May-07	Second Issue	Signed below	Signed below
A2	18-Nov-06	First Issue	Signed below	Signed below
MARK	DATE	DESCRIPTION	RAIL	CIVIL
APPROVED		Project Director	S. Shimamura N/A	
		Deputy Project Director	S. Sasaki 	
			Checked By (Project Director)	Y. Hiraoka 
			Checked By (QA/QC Manager)	J. Whorlow 
Checked By	N/A		Checked By (Project Manager)	N/A 
Checked By	N/A		Checked By (Engineering Manager)	M. Makino 
Prepared By	N/A		Issued By	B. Maney 
DATE	-		DATE	15 May 2007
RAIL SYSTEM			CIVIL JV	
CONTRACTOR'S DOCUMENT No.:		DOCUMENT No.:		REVISION
		DM001-E-AA0-CDP-DR-DCC-373904		A3

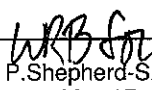
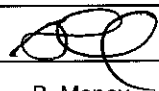
**Dubai Metro, UAE
Jebel Ali Depot**

**Structural Design Calculation
(Enhanced Design)
Building 11 – Electrical Store**

May 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-E-AAO-CDP-DR-DCC-373904	Controlled Copy No.:
Document Title: Auxiliary Depot- Jebel Ali Structural Design Calculation (Enhanced Design) Building 11 – Electrical Store	
Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 02 May 2007

Revision, Review and Approval/Authorization Records

A3	Issued for Approval (ENHANCED DESIGN)	 CK Lam May 07	 P. Shepherd-Smith May 07	 B. Maney May 07
A2	Issued for Approval	CK Lam Oct 06	P. Shepherd-Smith Oct 06	B. Maney Oct 06
A1	Internal Review	CK Lam Sep 06	P. Shepherd-Smith Sep 06	B. Maney Sep 06
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

Distribution (if insufficient space, please use separate paper)

Controlled Copy No.	Issued to
1	JT Metro JV
2	Atkins (Office Copy)
3	Atkins (P. Shepherd-Smith)
4	Meinhardt Infrastructure (CK Lam)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

Table of Contents

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2	FOUNDATION DESIGN CHECK	032
3	CONNECTIONS DESIGN	037

1 TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS

Document No.: **DM001-E-AAO-CDP-DR-DCC-373904**

Rev A3

Date: 02 May 2007

Page



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Job No

Sheet No

1

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By

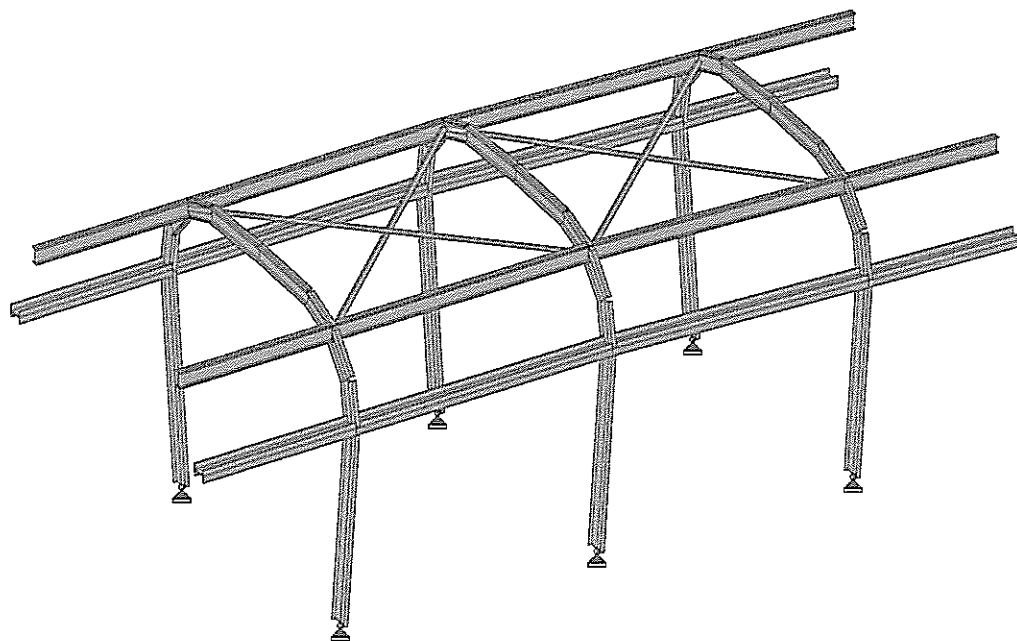
Date April 2007

Chd

Client

File Electrical Store.std

Date/Time 28-Apr-2007 15:18



Load 1

ELECTRICAL STORE

(JESSEL ALI - BUILDING II)



Site:	Dubai Metro Project	Date:	27-Apr-07	Job No.:	68010.01
	Electrical Store				
	Loadings	Designed by:	Rudy	Sheet No.:	

Roof Loadings :**DL** W_{DL} :

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
M&E Finishes	0.3 kPa		
	<u>0.8 kPa</u>	x 1 =	0.8 kN/m

LL

0.6 kPa	x 1 =	0.6 kN/m
---------	-------	----------

P_{LL} : A/C equipment S/W
 = 5 kN

(point load at mid-span of bay 2, 4)

Notional horizontal load, Ph

(Load Case of 1.4DL + 1.6LL)

Tributary load area for the roof

$$= (10+2*3)*(5+2*1.5)$$

$$= 128 \text{ m}^2$$

Total Dead Load

102.4 kN

Total Live Load

81.8 kN

(Including A/C equipment)

Tributary load area for the wall

$$= 6*2*(10+2*3+5+2*1.5)$$

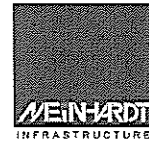
$$= 288 \text{ m}^2$$

Total Dead Load

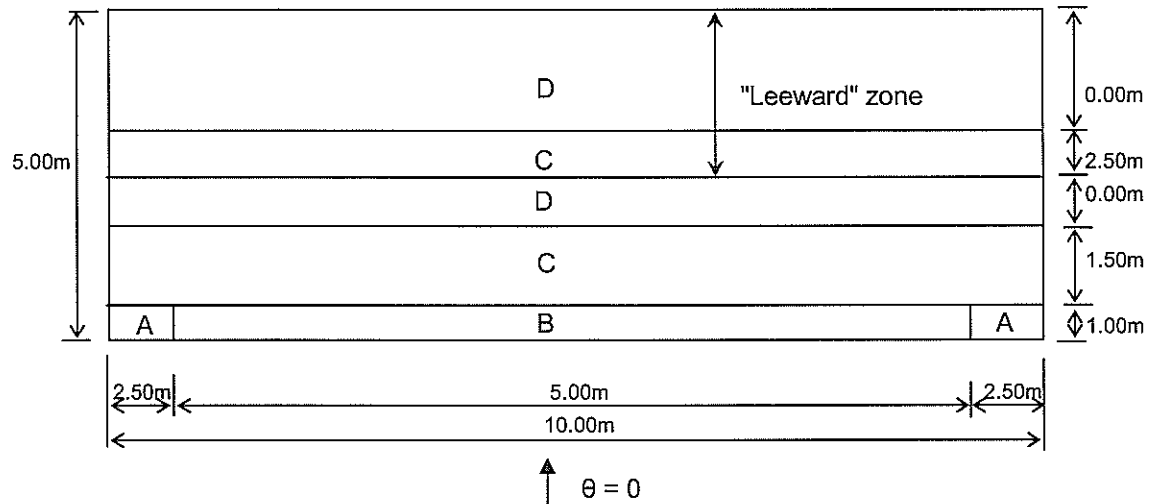
144 kN

Total Live Load

(excluding the M&E Finishes)



Site:	Dubai Metro Project	Date:	27-Apr-07	Job No.:	68010.01
	Electrical Store				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

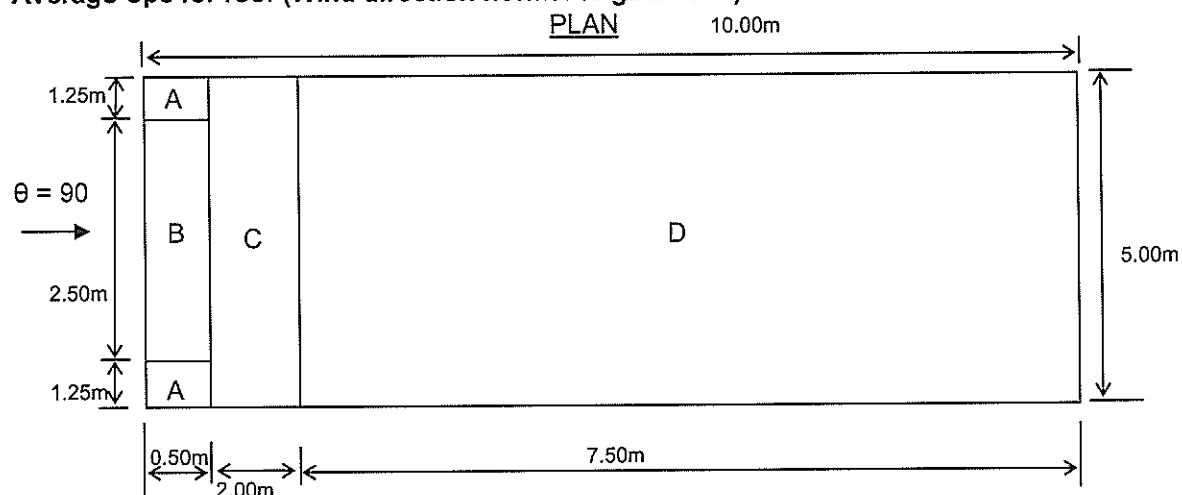
Average Cpe for roof (Wind direction normal to eave)
PLAN


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 10 \text{ m} \\
 H &= 6 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 10 \\
 &= 10 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	5.0	-2			
B	5.0	-1.4			
C	15.0	-0.7	25.0	-0.7	-0.7
D	0.0	-0.2	0.0	-0.2	0.2
Total	25.0		25.0		
Average		-1.10		-0.70	-0.70

Site:	Dubai Metro Project	Date:	27-Apr-07	Job No.:	68010.01
	Electrical Store				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 5 \text{ m} \\
 H &= 6 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 5 \\
 &= 5 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

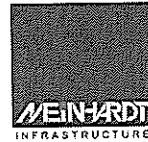
For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

Wind Load Computation

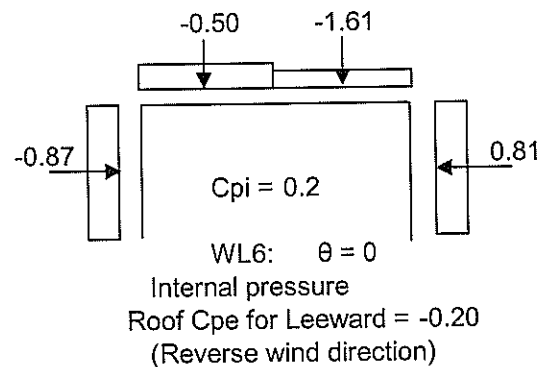
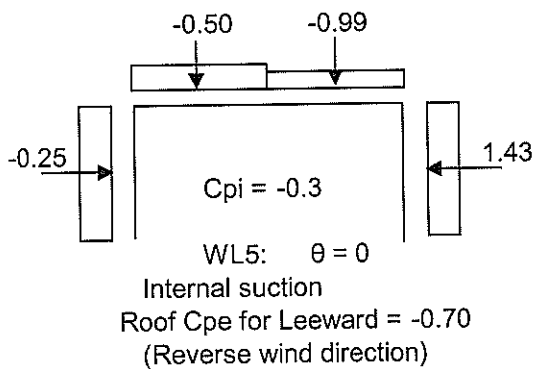
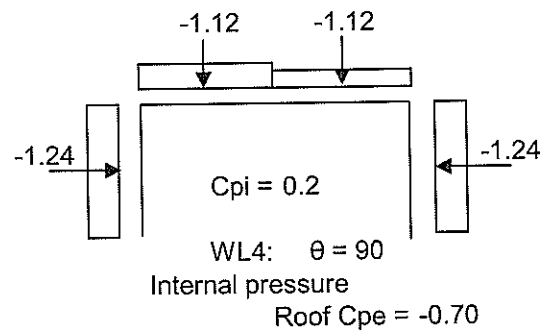
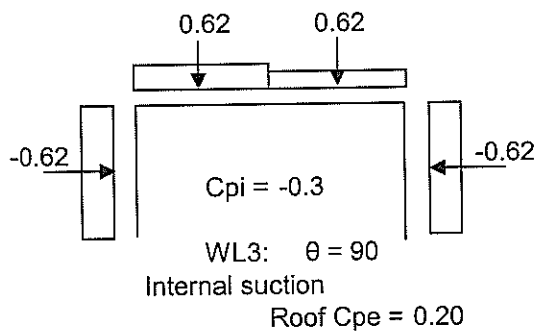
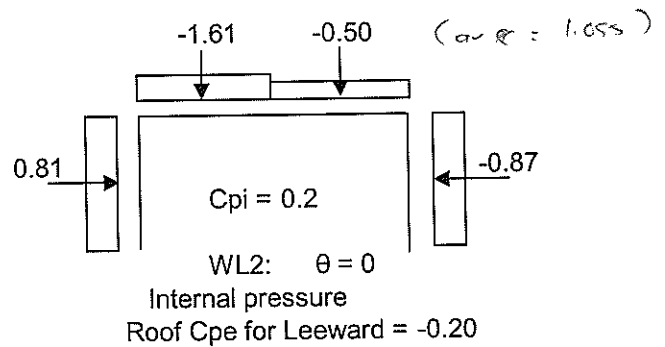
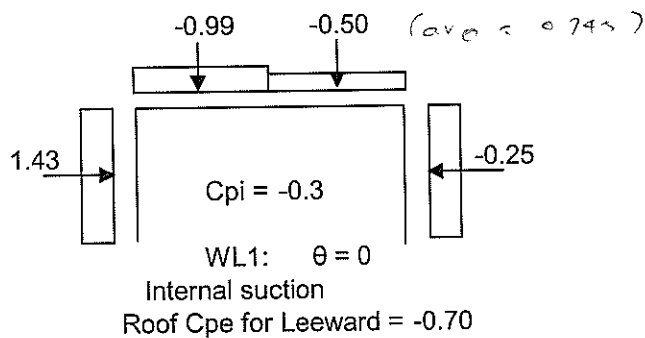
$$\begin{aligned}
 D &= 5.00 \text{ m} \\
 H &= 6 \text{ m} \\
 D/H &= 0.83
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-1.10	-0.70 -0.70	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.85	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)



Site:	Dubai Metro Project	Date:	27-Apr-07	Job No.:	68010.01
	Electrical Store				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



Site: Dubai Metro Project
Electrical Store
Earthquake Loading

Date: 27-Apr-07

Job No.: 68010.01

Designed by: Rudy

Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Table 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
Importance Factor $I = 1.00$ (Table 16-K)
Numeric. Coef. Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 35.11$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 6.95$

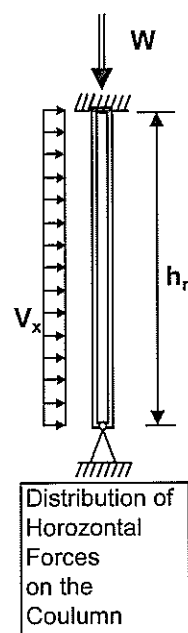
Design Earthquake Loads

- (Section 1630.6)
 $E_x = p \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h = \text{Earthquake load in the element due to the base shear, } V, \text{ as set out above (for non-structural components refer to section 1632)}$
Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
Dead Load on the element $W = D = \text{Given below}$
 $p = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq p \leq 1.5$)
 $r_{max} = \text{The maximum element-storey shear ratio}$
 $A_B = \text{Ground floor area of the structure}$

Distribution of Horizontal Forces:Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W, kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	$V_{base-S-Max}$ kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
C1	287	6.00	0.33	62.5	35.11	31.603	75.5	98.3	12.6	16.4

Load per column = $6 \times 12.6 / 6$
= 12.6 kN





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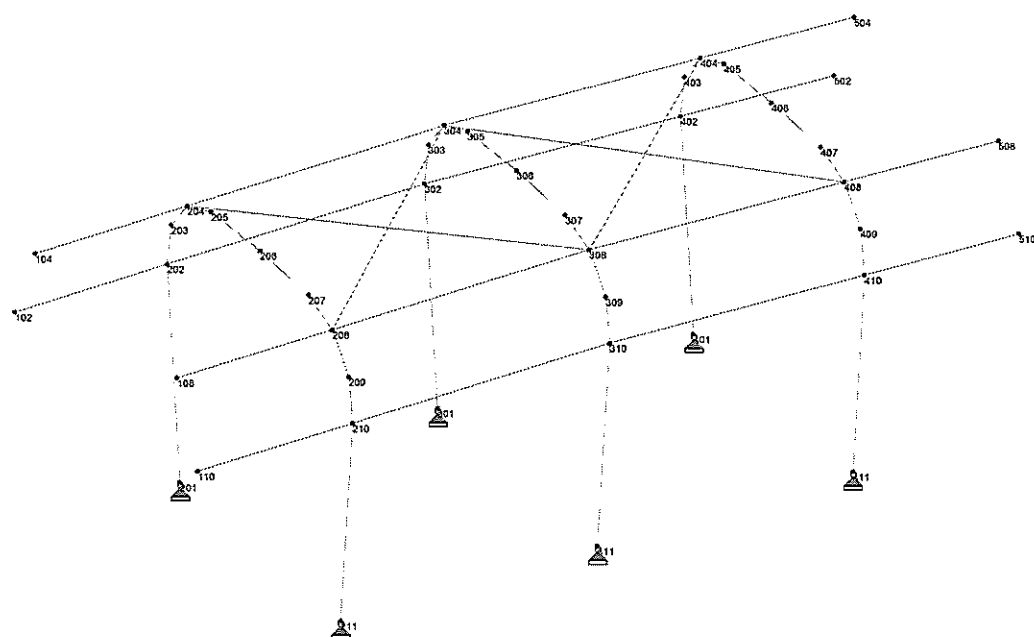
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Load 1



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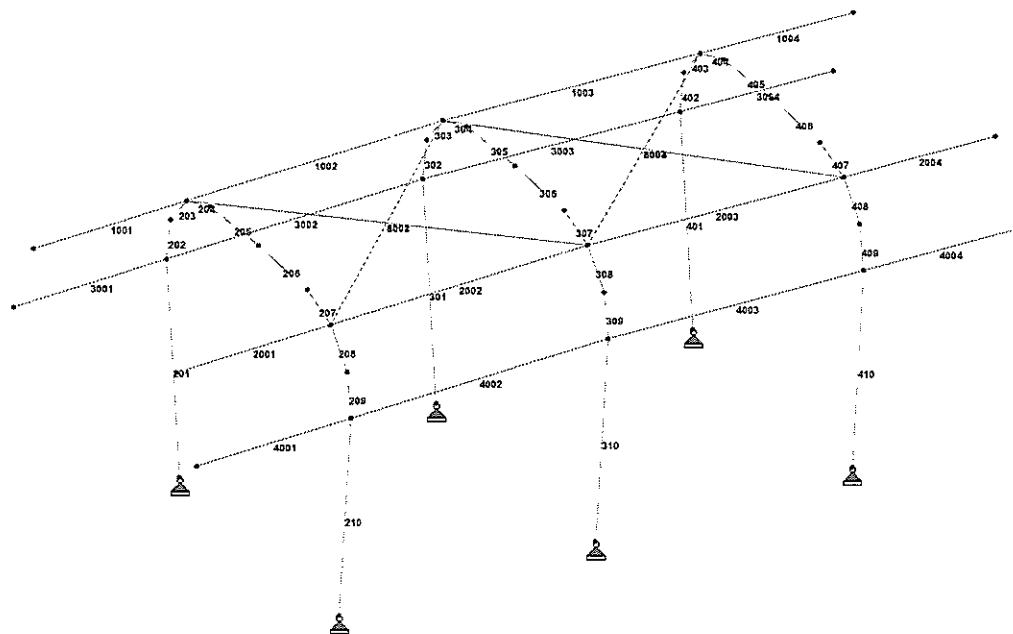
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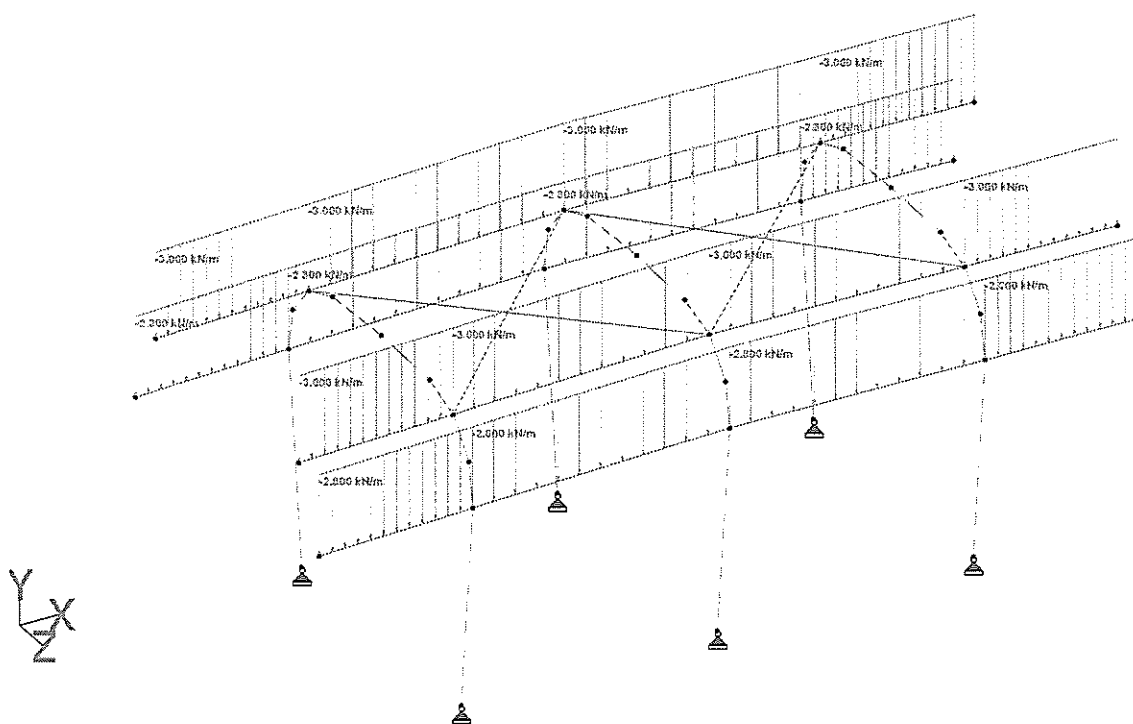
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Load 2



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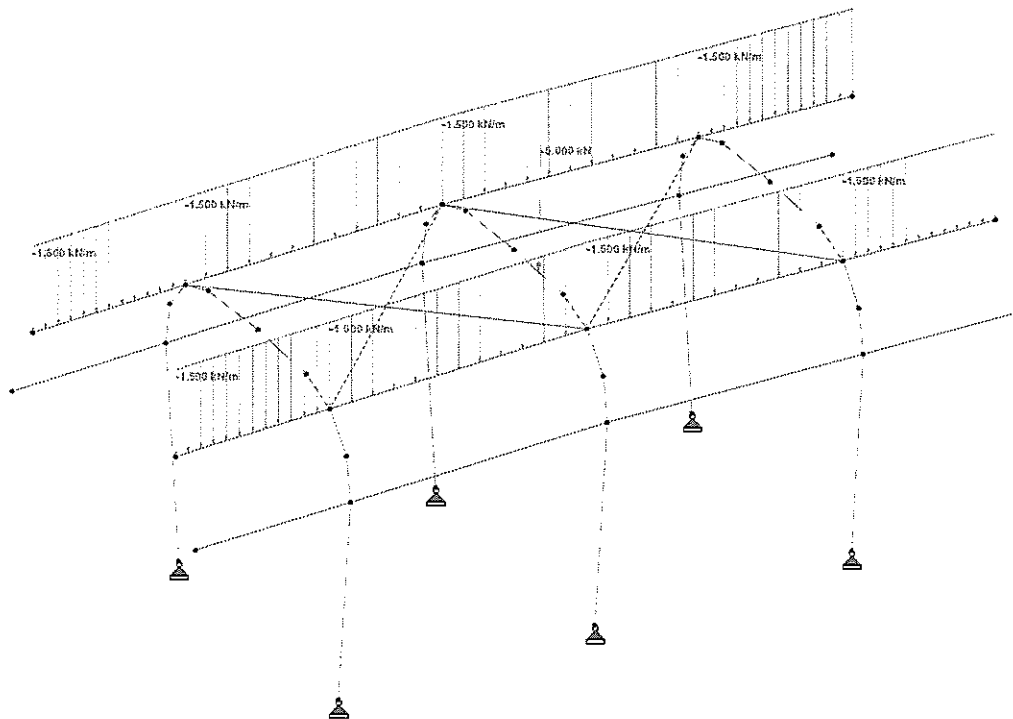
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Load 3



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Load 11



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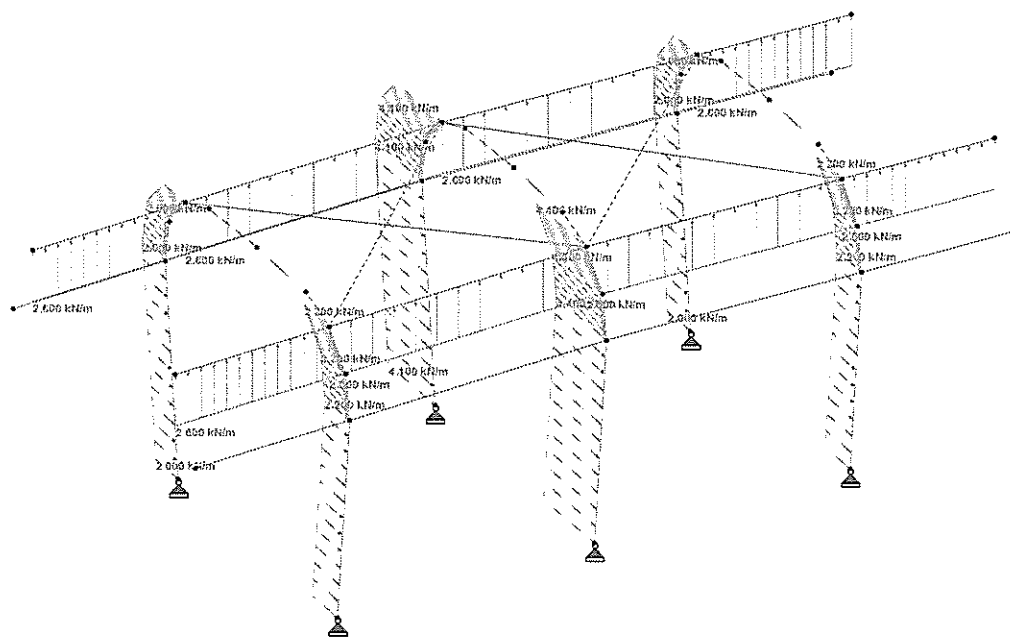
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Load 12



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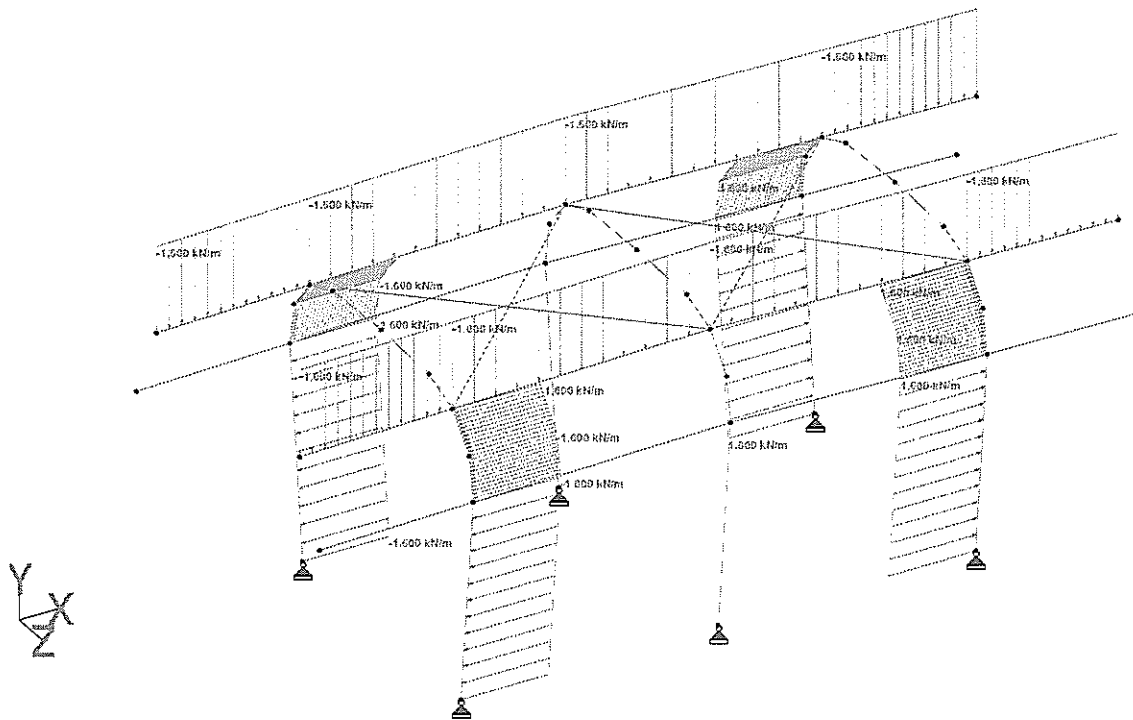
Date April 2007

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File Electrical Store.std

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Load 13



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Job No

Sheet No

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Job Title

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By

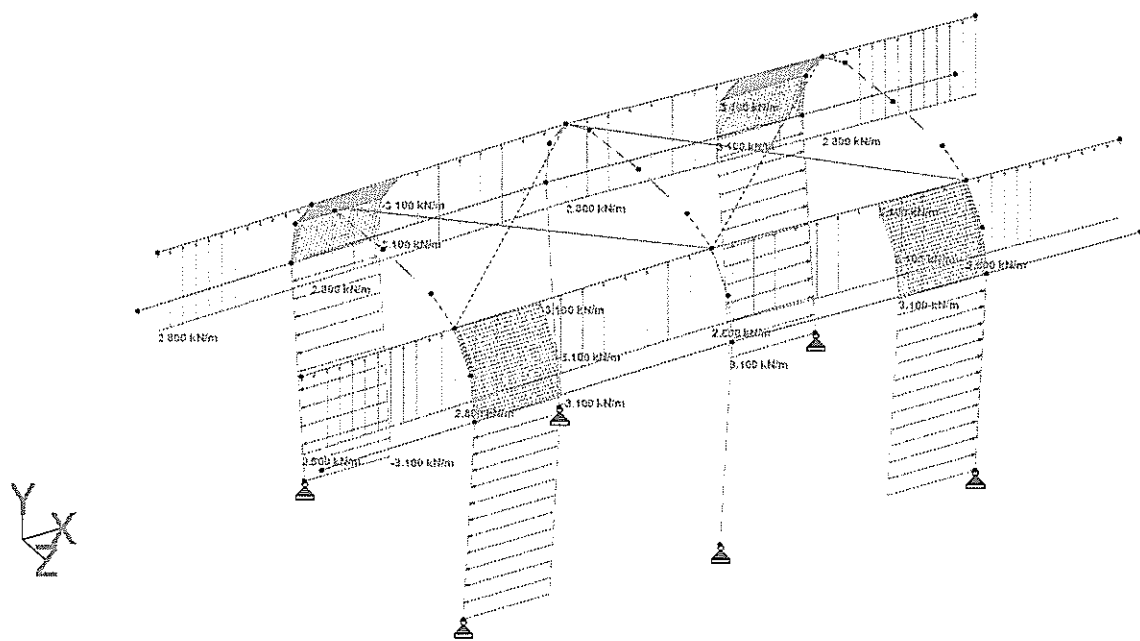
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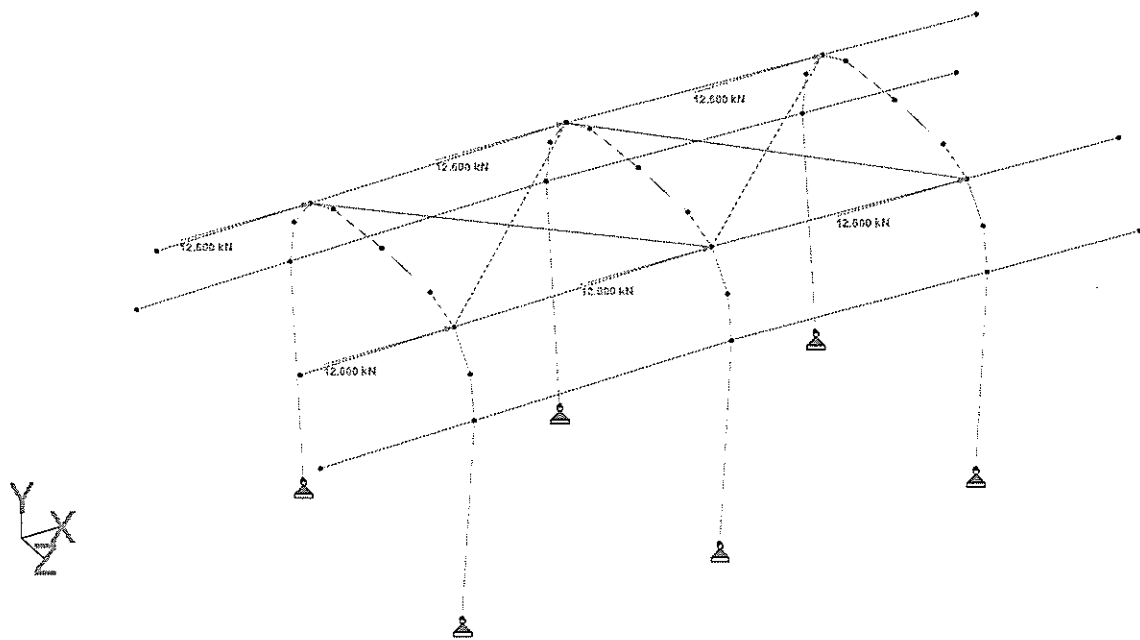


Load 14



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Job No	Sheet No 1	Rev
Part		
Job Title	Ref	
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Client	File Electrical Store.std	Date/Time 28-Apr-2007 15:18



Load 21



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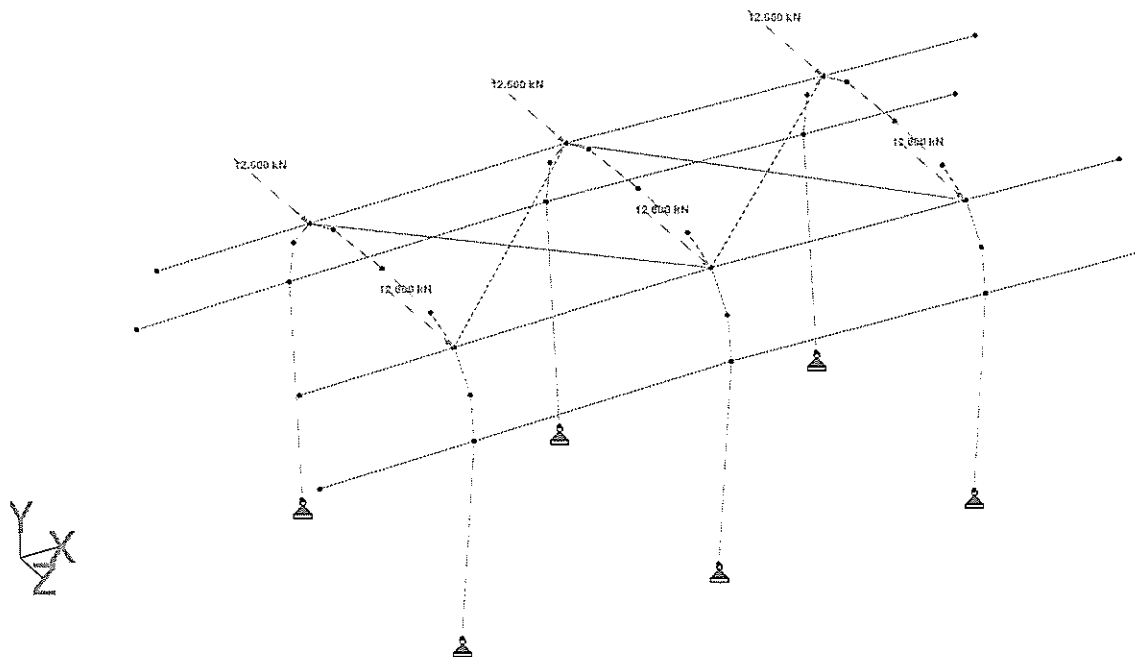
Date April 2007

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Client

File Electrical Store.std

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Load 22

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* Version 2006
* Proprietary Program of
* Research Engineers, Intl.
* Date= APR 28, 2007
* Time= 15:20:39
*
* USER ID: Weinhardt Infrastructure Pte Ltd
* *****

1. STAAD SPACE

INPUT FILE: Electrical Store.STD

2. START JOB INFORMATION

3. ENGINEER DATE APRIL 2007

4. END JOB INFORMATION

5. INPUT WIDTH 79

6. UNIT METER KN

7. JOINT COORDINATES

8. 102 -8.000 3.982 -3.102; 104 -8.000 5.464 -2.422
9. 110 -8.000 3.982 3.102; 502 8.000 3.982 -3.102
10. 504 8.000 5.464 -2.422; 108 -8.000 5.464 2.422
11. 508 8.000 5.464 2.422; 510 8.000 3.982 3.102
13. 201 -5.000 0.000 -2.685; 202 -5.000 4.054 -3.111
14. 203 -5.000 4.889 -2.974; 204 -5.000 5.538 -2.429
15. 205 -5.000 5.818 -1.631; 206 -5.000 5.875 0.000
16. 207 -5.000 6.888 2.931; 208 -5.000 7.538 2.429
17. 209 -5.000 8.889 2.974; 210 -5.000 9.538 3.111
18. 301 0.000 0.000 -2.685; 302 0.000 4.174 -3.124
19. 303 0.000 0.000 -2.685; 304 0.000 5.662 -2.440
21. 305 0.000 5.011 -2.986; 306 0.000 6.000 0.000
22. 307 0.000 5.943 -1.639; 308 0.000 5.662 2.440
23. 309 0.000 5.943 1.639; 310 0.000 5.662 2.440
24. 309 0.000 5.011 2.986; 310 0.000 4.174 3.124
25. 311 0.000 0.000 2.685
27. 401 5.000 0.000 -2.685; 402 5.000 4.054 -3.111
28. 403 5.000 4.889 -2.974; 404 5.000 5.538 -2.429
29. 405 5.000 5.818 -1.631; 406 5.000 5.875 0.000
30. 407 5.000 6.888 2.931; 408 5.000 7.538 2.429
31. 409 5.000 8.889 2.974; 410 5.000 9.538 3.111
32. 411 5.000 0.000 2.685
35. MEMBER INCIDENCES
36. 201 202 210; 301 302 310; 401 402 410
37. 1001 104 204; 1002 204 304; 1003 304 404; 1004 404 504
38. 2001 108 208; 2002 208 308; 2003 308 408; 2004 408 508
39. 3001 102 202; 3002 202 302; 3003 302 402; 3004 402 502
40. 4001 110 210; 4002 210 310; 4003 310 410; 4004 410 510
41. 8001 204 308; 8002 208 304; 8003 304 408; 8004 308 404
42. 4001 110 210; 4002 210 310; 4003 310 410; 4004 410 510
43. 8001 204 308; 8002 208 304; 8003 304 408; 8004 308 404
44. DEFINE MATERIAL START
STAAD SPACE

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45. ISOTROPIC STEEL
46. E 2.05E+008
47. POISSON 0.3
48. DENSITY 78.8195
49. ALPHA 1.2E-005
50. THERMAL 1.2E-005
51. END DEFINE MATERIAL
52. MEMBER PROPERTY BRITISH
53. 201 TO 210 301 TO 310 401 TO 410 -
54. 1002 1003 2002 2003 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
55. 1002 1003 2002 2003 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
56. 1001 1004 2001 2004 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
57. 3001 TO 3004 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
58. 4001 TO 4004 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
59. 8001 TO 8004 TABLE ST UA9DX90X6
60. CONSTANTS
61. BETA 90 MEMB 3001 TO 3004 4001 TO 4004
62. MATERIAL STEEL ALL
63. MEMBER TENSION
64. 8001 TO 8004
65. SUPPORTS
66. 201 211 301 411 PINNED
67. 301 411 SELFWEIGHT
68. 500 HEIGHT Y -1
69. LOAD 2 SLL
70. MEMBER LOAD

71. 1001 TO 1004 2001 TO 2004 UNI GY -3.0
72. 3001 TO 3004 4001 TO 4004 UNI GY -2.8
73. LOAD 3 SLL
74. MEMBER LOAD
75. 1001 TO 1004 2001 TO 2004 UNI GY -1.5
76. 306 CON Y -5
77. LOAD 11 WLL (INTERNAL SUCTION, WIND ANGLE 0)
78. MEMBER LOAD
79. 1001 TO 1004 2001 TO 2004 UNI GY 1.9
80. 201 TO 203 401 TO 403 UNI GZ 3.6
81. 301 TO 303 UNI GZ 7.2
82. 208 TO 210 408 TO 410 UNI GZ 0.6
83. 308 TO 310 UNI GZ 1.2
85. LOAD 12 WLL (INTERNAL PRESSURE, WIND ANGLE 0)
86. MEMBER LOAD
87. 1001 TO 1004 2001 TO 2004 UNI GY 2.6
88. 301 TO 303 401 TO 403 UNI GZ 2.0
89. 301 TO 303 UNI GZ 4.1
90. 208 TO 210 408 TO 410 UNI GZ 2.2
91. 308 TO 310 UNI GZ 4.4
93. LOAD 13 WLL (INTERNAL SUCTION, WIND ANGLE 90)
94. MEMBER LOAD
95. 1001 TO 1004 2001 TO 2004 UNI GY -1.6
96. 201 TO 203 208 TO 210 UNI GY -1.6
97. 401 TO 403 408 TO 410 UNI GY 1.6
99. LOAD 14 WLL (INTERNAL PRESSURE, WIND ANGLE 90)
100. MEMBER LOAD
101. 1001 TO 1004 2001 TO 2004 UNI GY 2.8
102. 201 TO 203 208 TO 210 UNI GY -3.1
103. 401 TO 403 408 TO 410 UNI GY 3.1
106. STAAD SPACE
107. JOINT LOAD
108. 204 208 304 308 404 408 FX 12.6
109. LOAD 22 EQ1 (EARTHQUAKE LOAD)
110. JOINT LOAD
111. 204 208 304 308 404 408 FZ 12.6
113. LOAD COMB 1001 1.4DL + 1.6LL
114. 1.1 2 1.4 3 1.6
116. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
117. 1.1 2 1.2 3 1.2 11 1.2
118. LOAD COMB 1111 1.4DL + 1.4WL1
119. 1.1 2 1.4 11 1.4
120. LOAD COMB 1211 1.0DL + 1.4WL1
121. 1.1 2 1.0 11 1.4
122. LOAD COMB 2012 1.2DL + 1.2LL + 1.2WL2
123. 1.1 2 2.0 12 1.2 1.1
124. LOAD COMB 1112 1.4DL + 1.4WL2
125. 1.1 2 1.4 12 1.4
126. LOAD COMB 1212 1.0DL + 1.4WL2
127. 1.1 2 1.0 12 1.4
128. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
129. 1.1 2 1.2 3 1.2 13 1.2
130. LOAD COMB 1113 1.4DL + 1.4WL3
131. 1.1 2 1.4 13 1.4
132. LOAD COMB 1213 1.0DL + 1.4WL3
133. 1.1 2 1.0 13 1.4
134. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
135. 1.1 2 1.2 3 1.2 14 1.2
136. LOAD COMB 1114 1.4DL + 1.4WL4
137. 1.1 2 1.4 14 1.4
138. LOAD COMB 1214 1.0DL + 1.4WL4
139. 1.1 2 1.0 14 1.4
140. LOAD COMB 1015 1.2DL + 1.2LL + 1.0EQ1
141. 1.1 2 1.2 3 0.5 21 1.0
142. 1.1 2 1.2 3 0.5 22 1.0
143. LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
144. 1.1 2 1.2 3 0.5 22 1.0
145. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ1
146. 1.1 2 1.2 3 0.5 21 -1.0
147. LOAD COMB 1122 1.2DL + 0.5LL - 1.0EQ2
148. 1.1 2 1.2 3 0.5 22 -1.0
149. LOAD COMB 1221 0.9DL + 1.0EQ1
150. 1.0 2 0.9 21 1.0
151. LOAD COMB 1222 0.9DL + 1.0EQ2
152. 1.0 2 0.9 22 1.0
153. LOAD COMB 1321 0.9DL - 1.0EQ1
154. 1.0 2 0.9 21 -1.0
155. LOAD COMB 1322 0.9DL - 1.0EQ2
156. 1.0 2 0.9 22 -1.0
157. *SLS
158. LOAD COMB 2001 1.0DL + 1.0LL
159. 1.1 2 1.0 3 1.0
160. LOAD COMB 2002 1.0DL + 0.5LL
161. 1.1 2 1.0 3 0.5
162. LOAD COMB 2011 DL + LL + WL1

Electrical Store.ANL

163. 1 1.0 2 1.0 3 1.0 11 1.0
164. LOAD COMB 2012 DL + LL + WL2
165. 1 1.0 2 1.0 3 1.0 12 1.0
STAAD SPACE
-- PAGE NO. 4

166. LOAD COMB 2013 DL + LL + WL3
167. 1 1.0 2 1.0 3 1.0 13 1.0
168. LOAD COMB 2014 DL + LL + WL4
169. 1 1.0 2 1.0 3 1.0 14 1.0
171. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBERELEMENTS/SUPPORTS = 41/ 50/ 6
ORIGINAL/FINAL BAND-WIDTH = 32/ 11/
TOTAL DEGREES OF FREEDOM = 228
SIZE OF STIFFNESS MATRIX = 9, 17 DOUBLE KILO-WORDS
REQD/AVAIL. DISK SPACE = 12.3/ 416818.9 MB

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**NOTE-Tension/Compression converged after 1 iterations, Case= 2

**NOTE-Tension/Compression converged after 1 iterations, Case= 3

**START ITERATION NO. 2

**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 11

**START ITERATION NO. 2

**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 12

**NOTE-Tension/Compression converged after 1 iterations, Case= 13

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 14

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 21

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 22

172. LOAD LIST 1001 1011 TO 1014 1021 1022 1111 TO 1114 1121 1122 -
173. 1211 TO 1214 1221 1222 1321 1322

174. PRINT SUPPORT REACTION ALL

SUPPORT REACTION ALL

STAAD SPACE
-- PAGE NO. 6

Page 3

Electrical Store.ANL

STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
201	1001	-0.23	77.43	-2.94	0.00	0.00	0.00
	1011	-0.20	28.34	-25.27	0.00	0.00	0.00
	1111	-0.24	20.04	-29.44	0.00	0.00	0.00
	1211	-0.17	23.17	-28.61	0.00	0.00	0.00
	1012	-0.21	23.15	-29.46	0.00	0.00	0.00
	1112	-0.18	23.77	-23.64	0.00	0.00	0.00
	1212	-0.18	23.77	-23.64	0.00	0.00	0.00
	1013	2.81	75.56	-2.52	0.00	0.00	0.00
	1113	3.27	75.13	-2.90	0.00	0.00	0.00
	1213	3.34	57.26	-2.07	0.00	0.00	0.00
	1014	5.60	42.45	-2.23	0.00	0.00	0.00
	1114	6.53	36.51	-2.56	0.00	0.00	0.00
	1214	6.60	18.64	-1.73	0.00	0.00	0.00
	1021	-12.79	37.17	-0.75	0.00	0.00	0.00
	1022	-0.21	31.52	-15.29	0.00	0.00	0.00
	1121	12.38	79.36	-4.24	0.00	0.00	0.00
	1122	-0.20	85.01	10.31	0.00	0.00	0.00
	1221	-12.74	19.12	-0.11	0.00	0.00	0.00
	1222	-0.16	13.46	-14.66	0.00	0.00	0.00
	1321	12.43	61.30	-3.61	0.00	0.00	0.00
	1322	-0.13	95.96	10.94	0.00	0.00	0.00
211	1001	-0.21	77.28	-3.43	0.00	0.00	0.00
	1011	-0.21	77.28	-3.43	0.00	0.00	0.00
	1111	-0.18	54.07	-13.11	0.00	0.00	0.00
	1211	-0.18	54.07	-13.11	0.00	0.00	0.00
	1012	-0.22	68.01	-15.67	0.00	0.00	0.00
	1112	-0.26	66.49	-18.40	0.00	0.00	0.00
	1212	-0.19	48.62	-19.22	0.00	0.00	0.00
	1013	2.81	75.41	2.58	0.00	0.00	0.00
	1113	3.27	75.13	2.90	0.00	0.00	0.00
	1213	3.34	57.26	2.07	0.00	0.00	0.00
	1014	5.60	42.31	2.29	0.00	0.00	0.00
	1114	6.53	36.51	2.56	0.00	0.00	0.00
	1214	6.60	18.64	1.73	0.00	0.00	0.00
	1021	-12.79	37.11	0.77	0.00	0.00	0.00
	1022	-0.20	84.94	-10.23	0.00	0.00	0.00
	1121	12.38	79.30	4.27	0.00	0.00	0.00
	1122	-0.21	17.17	15.27	0.00	0.00	0.00
	1221	-16.14	66.95	-10.80	0.00	0.00	0.00
	1222	-0.16	61.30	3.61	0.00	0.00	0.00
	1321	12.43	13.48	14.61	0.00	0.00	0.00
	1322	-0.16	53.79	-1.67	0.00	0.00	0.00
301	1001	0.00	53.79	-1.67	0.00	0.00	0.00
	1011	0.00	18.05	-33.76	0.00	0.00	0.00
	1111	0.00	10.79	-39.51	0.00	0.00	0.00
	1211	0.00	-1.22	-38.99	0.00	0.00	0.00
	1012	0.00	14.41	-24.46	0.00	0.00	0.00
	1112	0.00	6.54	-28.66	0.00	0.00	0.00
STAAD SPACE							
							-- PAGE NO. 7

Electrical Store.ANL

STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1212		0.00	-5.48	-28.14	0.00	0.00	0.00
1013		0.00	54.01	-1.71	0.00	0.00	0.00
1113		0.00	52.75	-2.12	0.00	0.00	0.00
1213		0.00	40.73	-1.60	0.00	0.00	0.00
1014		0.00	35.71	-1.98	0.00	0.00	0.00
1114		0.00	31.40	-2.43	0.00	0.00	0.00
1214		0.00	19.38	-1.91	0.00	0.00	0.00
1021		-12.64	39.72	-1.53	0.00	0.00	0.00
1022		0.00	14.66	-13.80	0.00	0.00	0.00
1121		12.64	39.71	-1.49	0.00	0.00	0.00
1122		0.00	27.66	10.18	0.00	0.00	0.00
1221		-12.64	1.98	-13.45	0.00	0.00	0.00
1222		0.00	27.03	11.14	0.00	0.00	0.00
1321		12.64	52.08	11.13	0.00	0.00	0.00
1322		0.00	55.99	1.78	0.00	0.00	0.00
1001		0.00	57.09	-12.18	0.00	0.00	0.00
1111		0.00	54.41	-14.18	0.00	0.00	0.00
1211		0.00	42.39	-14.70	0.00	0.00	0.00
1012		0.00	53.36	-21.97	0.00	0.00	0.00
1112		0.00	50.06	-25.61	0.00	0.00	0.00
STAAD SPACE							
							Page 4

Electrical Store-AWL										
201	0.00	MAX	19.13	1122	0.00	1022	12.43	1321	0.00	1321
	0.82	MIN	-28.23	1211	0.00	1211	-12.79	1021	0.00	1021
	0.82	MAX	19.10	1122	21.36	1211	12.43	1321	10.13	1321
	1.63	MIN	-24.17	1211	-15.59	1122	-12.79	1021	-10.42	1021
	2.45	MAX	19.07	1122	39.40	1211	12.43	1321	-20.85	1021
	3.26	MIN	-20.11	1211	-31.14	1122	-12.79	1021	-30.40	1321
	4.08	MAX	19.04	1122	54.14	1211	12.43	1321	-41.57	1321
		MIN	-16.18	1212	-46.66	1122	-12.79	1021	-41.57	1321
		MAX	13.00	1122	65.17	1211	12.43	1321	-40.70	1021
		MIN	-18.97	1122	-73.66	1211	-12.79	1021	-50.66	1321
		MAX	13.29	1222	-77.66	1122	-12.79	1021	-52.12	1021

MAX/MIN FORCE VALUES FOR MEMB										
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	
MAX.	19.13	0.00	1122	73.69	4.08	1211				
MIN.	-28.23	0.00	1211	-50.66	4.08	1321	83.47	C	0.00	
	-12.79	4.08	1021	-52.12	4.08	1021	3.85	T	4.08	

202	0.00	MAX	15.87	1022	77.64	1122	11.12	1021	5.89	1321
	0.17	MIN	-4.39	1322	-73.69	1211	-7.30	1321	-9.19	1021
	0.34	MAX	15.86	1022	77.99	1122	11.12	1021	4.66	1321
	0.51	MIN	-4.40	1322	-74.05	1211	-7.30	1321	-7.31	1021
	0.68	MAX	15.85	1022	78.34	1122	11.12	1021	3.42	1321
	0.85	MIN	-4.40	1322	-74.26	1211	-7.30	1321	-5.43	1021
		MAX	15.84	1022	78.70	1122	11.12	1021	2.19	1321
		MIN	-4.41	1322	-74.33	1211	-7.30	1321	-3.55	1021
		MAX	15.83	1022	79.06	1122	11.12	1021	0.95	1321
		MIN	-4.42	1322	-74.26	1211	-7.30	1321	-0.95	1021
		MAX	15.82	1022	79.42	1122	11.12	1021	-0.75	1321
		MIN	-4.43	1322	-74.66	1212	-7.30	1321	-0.75	1021

STAAD SPACE										

MAX/MIN FORCE VALUES FOR MEMB										
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	
MAX.	15.87	0.00	1022	79.42	0.85	1122				
MIN.	-4.43	0.85	1322	-74.66	0.85	1321	58.64	C	0.00	
	-7.30	0.85	1321	-9.19	0.00	1021	28.15	T	0.00	

203	0.00	MAX	32.81	1013	79.42	1122	11.12	1021	0.20	1321
	0.17	MIN	-12.19	1211	-74.66	1212	-7.30	1321	-0.27	1121
	0.34	MAX	32.77	1013	74.68	1122	11.12	1021	2.09	1021
	0.51	MIN	-12.68	1211	-72.74	1212	-7.30	1321	-1.50	1321
	0.68	MAX	32.73	1013	69.96	1122	11.12	1021	3.97	1021
	0.85	MIN	-13.57	1211	-70.75	1212	-7.30	1321	-2.74	1321
		MAX	32.68	1013	65.98	1122	11.12	1021	5.86	1021
		MIN	-14.26	1211	-69.49	1022	-7.30	1321	-7.78	1321
		MAX	32.64	1013	62.54	1122	11.12	1021	5.74	1021
		MIN	-14.84	1211	-71.97	1212	-7.30	1321	-9.63	1021
		MAX	32.60	1013	62.54	1122	11.12	1021	9.63	1321
		MIN	-15.03	1211	-74.24	1022	-7.30	1321	-6.45	1321

MAX/MIN FORCE VALUES FOR MEMB										
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	
MAX.	32.81	0.00	1013	79.42	0.00	1122				
MIN.	-15.63	0.85	1211	-74.66	0.00	1212	51.23	C	0.00	
	-7.30	0.85	1321	-6.45	0.85	1321	25.98	T	0.00	

204	0.00	MAX	24.75	1322	59.08	1322	0.00	1211	0.26	1021
	0.17	MIN	-28.67	1111	-74.27	1022	0.00	1022	-0.13	1321
		MAX	24.71	1322	54.90	1322	0.00	1211	0.26	1021
		MIN	-28.75	1111	-69.71	1022	0.00	1022	-0.13	1321

175. PRINT FORCE ENVELOPE NSECTION 5 ALL FORCE ENVELOPE NSECTION 5 STAAD SPACE

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN

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Electrical Store ANL

207, AMONGST ALL SECT LOCATIONS

MAX/MIN FORCE VALUES FOR MEMB

208

209

210

211

212

213

214

215

216

217

218

219

220

Electrical Store ANL

204, AMONGST ALL SECT LOCATIONS

MAX/MIN FORCE VALUES FOR MEMB

205

206

207

208

209

210

211

212

213

214

215

216

217

Electrical Store.AML

MAX/MIN FORCE VALUES FOR MEMB										210, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD					
MAX.	MIN.	FY	FZ	DIST	LD	DIST	LD	FX	DIST	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD
13.24	12.79	0.00	0.00	1322	0.00	1322	0.00	1322	83.41 C	4.08	1022								
-25.25	-25.25	4.08	1112	-77.32	0.00	1022	13.78 C	0.00	1322										
-12.43	-12.43	4.08	1321	-52.12	0.00	1021													
0.00	0.00	MAX	17.50	1122	0.00	1322	12.64	1321	0.00	1221									
0.84	0.84	MIN	-38.91	1211	0.00	1112	-12.64	1221	0.00	1121									
1.68	1.68	MIN	-30.52	1211	0.00	1211	-12.64	1221	10.61	1321									
2.52	2.52	MIN	-17.43	1211	0.00	1211	-12.64	1221	-10.61	1321									
3.36	3.36	MIN	-17.43	1211	0.00	1211	-12.64	1221	-21.21	1321									
4.20	4.20	MIN	-17.43	1211	0.00	1211	-12.64	1221	-31.82	1321									
		STAAD SPACE																	

MAX/MIN FORCE VALUES FOR MEMB

MAX/MIN FORCE VALUES FOR MEMB										301, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD					
MAX.	MIN.	FY	FZ	DIST	LD	DIST	LD	FX	DIST	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD
17.50	12.64	0.00	0.00	1122	75.50	4.20	1211	63.28 C	0.00	1122									
-38.91	-38.91	0.00	0.00	1211	-73.09	4.20	1121	6.34 T	4.20	1212									
-12.64	-12.64	4.20	1221	-53.03	4.20	1221													
0.00	0.00	MAX	12.74	1022	72.78	1122	20.04	1221	17.06	1321									
0.17	0.17	MIN	-2.58	1322	-75.84	1211	-20.04	1321	-17.06	1321									
0.34	0.34	MIN	-3.53	1211	-72.90	1122	-20.04	1321	13.66	1321									
0.51	0.51	MIN	-5.23	1211	-73.02	1122	-20.04	1321	-13.66	1321									
0.68	0.68	MIN	-5.23	1211	-73.02	1122	-20.04	1321	-10.26	1321									
0.85	0.85	MIN	-5.23	1211	-73.02	1122	-20.04	1321	-10.26	1321									
		STAAD SPACE																	

MAX/MIN FORCE VALUES FOR MEMB

MAX/MIN FORCE VALUES FOR MEMB										302, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD					
MAX.	MIN.	FY	FZ	DIST	LD	DIST	LD	FX	DIST	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD
12.74	12.74	0.00	0.00	1022	73.38	0.85	1122	46.98 C	0.00	1122									
-20.04	-20.04	0.00	0.00	1021	-17.06	0.00	1211	21.28 T	0.00	1212									
-10.32	-10.32	0.85	1211	-75.84	0.00	1211													
-20.04	-20.04	0.85	1321	-17.06	0.00	1321													
0.00	0.00	MAX	23.57	1013	73.38	1122	20.04	1221	0.07	1321									
0.17	0.17	MIN	-38.91	1211	-75.84	1211	-20.04	1321	-3.33	1321									
0.34	0.34	MIN	-17.43	1211	-75.84	1211	-20.04	1321	-6.74	1321									
0.51	0.51	MIN	-17.43	1211	-75.84	1211	-20.04	1321	-10.14	1321									
0.68	0.68	MIN	-17.43	1211	-75.84	1211	-20.04	1321	-13.55	1321									
0.85	0.85	MIN	-17.43	1211	-75.84	1211	-20.04	1321	-16.95	1321									
		STAAD SPACE																	

MAX/MIN FORCE VALUES FOR MEMB

MAX/MIN FORCE VALUES FOR MEMB										303, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD					
MAX.	MIN.	FY	FZ	DIST	LD	DIST	LD	FX	DIST	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD
12.74	12.74	0.00	0.00	1022	73.38	0.85	1122	46.98 C	0.00	1122									
-20.04	-20.04	0.00	0.00	1021	-17.06	0.00	1211	21.28 T	0.00	1212									
-10.32	-10.32	0.85	1211	-75.84	0.00	1211													
-20.04	-20.04	0.85	1321	-17.06	0.00	1321													
0.00	0.00	MAX	23.57	1013	73.38	1122	20.04	1221	0.07	1321									
0.17	0.17	MIN	-38.91	1211	-75.84	1211	-20.04	1321	-3.33	1321									
0.34	0.34	MIN	-17.43	1211	-75.84	1211	-20.04	1321	-6.74	1321									
0.51	0.51	MIN	-17.43	1211	-75.84	1211	-20.04	1321	-10.14	1321									
0.68	0.68	MIN	-17.43	1211	-75.84	1211	-20.04	1321	-13.55	1321									
0.85	0.85	MIN	-17.43	1211	-75.84	1211	-20.04	1321	-16.95	1321									
		STAAD SPACE																	

Electrical Store.AML

MAX/MIN FORCE VALUES FOR MEMB										210, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD					
MAX.	MIN.	FY	FZ	DIST	LD	DIST	LD	FX	DIST	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD
23.57	20.04	0.00	0.00	1013	73.38	0.00	1122	40.52 C	0.00	1122									
-33.51	-33.51	0.85	1211	-16.95	0.85	1321													
-20.04	-20.04	0.85	1321	-16.95	0.85	1321													
0.00	0.00	MAX	22.04	1322	56.07	1322	0.00	1321	0.26	1221									
0.17	0.17	MIN	-30.33	1111	-68.82	1022	0.00	1021	-0.26	1321									
0.34	0.34	MIN	-30.40	1111	-68.82	1022	0.00	1021	-0.26	1321									
0.51	0.51	MIN	-30.40	1111	-68.82	1022	0.00	1021	-0.26	1321									
0.68	0.68	MIN	-30.40	1111	-68.82	1022	0.00	1021	-0.26	1321									
0.85	0.85	MIN	-30.40	1111	-68.82	1022	0.00	1021	-0.26	1321									
		STAAD SPACE																	

MAX/MIN FORCE VALUES FOR MEMB

MAX/MIN FORCE VALUES FOR MEMB										304, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD				MZ/		DIST		LD					
MAX.	MIN.	FY	FZ	DIST	LD	DIST	LD	FX	DIST	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD
22.04	0.00	0.00	0.00	1322	56.07	0.00	1322	26.96 C	0.00	1011									
-30.68	-30.68	0.85	1111	-68.82	0.00	1022													
0.00	0.00	0.85	1021	-0.26	0.85	1321													
0.00	0.00	MAX	26.06	1122	37.46	1322	0.00	1221	0.28	1221									
0.33	0.33	MIN	-34.89	1322	-45.63	1022	0.00	1121	-0.28	1321									
0.66	0.66	MIN	-34.89	1322	-45.63	1022	0.00	1121	-0.28	1321									
0.98	0.98	MIN	-34.89	1322	-45.63	1022	0.00	1121	-0.28	1321									
1.31	1.31	MIN	-34.89	1322	-45.63	1022	0.00	1121	-0.28	1321									
1.64	1.64	MIN	-34.89	1322	-45.63	1022	0.00	1121	-0.28	1321									
		STAAD SPACE																	

MAX/MIN FORCE VALUES FOR MEMB

MAX/MIN FORCE VALUES FOR MEMB										305, AMONGST ALL SECT LOCATIONS		
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		Electrical Store-AMU									
		0.00		1.64		1.64		5.85		C	

2 STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB						406. AMONGST ALL SECT LOCATIONS							
	FZ	FX	LD	DIST	LD	MZ	FX	LD	DIST	LD	FX	DIST	LD
MAX.	26.96	0.00	0.00	1122		45.29	1.63	1211					
	0.00	0.00	0.00	1022		0.14	0.00	1221		18.55	C	1.63	1011
	-27.00	0.00	1.63	1022		-51.74	1.63	1122					
MIN.	0.00	1.63	1.63	1211		-0.27	0.00	1121		2.46	C	0.00	1322

#07	0.00	MAX	26.72	1122	45.29	1211	0.00	1022	0.13	1221
		MIN	-58.97	1222	-51.77	1211	0.00	1211	-0.26	1221
0.17		MAX	-56.67	1122	-48.63	1211	0.00	1022	0.13	1221
		MIN	-53.99	1222	-56.26	1122	0.00	1211	-0.26	1221
0.34		MAX	-56.61	1122	-51.98	1211	0.00	1022	0.13	1221
		MIN	-54.03	1222	-60.77	1122	0.00	1211	-0.26	1221
0.51		MAX	-56.55	1122	-55.33	1211	0.00	1022	0.13	1221
		MIN	-54.08	1222	-65.26	1122	0.00	1211	-0.26	1221
0.68		MAX	-56.49	1122	-58.69	1211	0.00	1022	0.13	1221
		MIN	-54.12	1222	-69.75	1122	0.00	1211	-0.26	1221
0.85		MAX	-56.43	1122	-62.07	1211	0.00	1022	0.13	1221
		MIN	-54.17	1222	-74.22	1122	0.00	1211	-0.26	1221

MAX/MIN FORCE VALUES FOR MEMB				407, AMONGST ALL SECT LOCATIONS									
	FY	FZ		LD	DIST	LD	DIST	LD	DIST	LD	DIST	LD	DIST
MAX.	26.73	0.00	0.00	1122		62.07	0.85	1211					
	0.00	0.00	0.00	1022		0.13	0.00	1221					
MIN.	-24.17	0.85	1222			-74.22	0.85	1122		24.67	C	0.85	1111
	0.00	0.85	1211			-0.76	0.00	1121		5.49		0.00	1322

[illegible]

	MAX/MIN FORCE VALUES FOR MEMB			408, AMONGST ALL SECT LOCATIONS					
	F _y /F _z	LD	DIST	LD	DIST	LD	DIST	LD	DIST
MAX.	0.35	0.00	1214	79.01	0.85	1022			
	11.10	0.00	1121	6.44	0.00	1121	51.11	C	0.85 1022
MIN.	-32.84	0.85	1013	-74.19	0.00	1122			
	-7.79	0.85	1321	-9.61	0.00	1121	15.47		0.00 1322

	MAX	4.41	1222	79.01	1022	11.10	1121	0.29	1021
0.00	MIN	-45.86	1222	-75.17	1322	-7.39	1221	-0.22	1321
0.17	MAX	4.40	1222	68.16	1022	11.10	1221	1.06	1221
	MIN	-45.87	1222	-62.88	1322	-7.39	1221	-0.85	1321
0.34	MAX	4.42	1211	78.32	1022	11.10	1221	0.95	1221
	MIN	-45.88	1222	-60.60	1322	-7.39	1221	-0.18	1321
0.51	MAX	4.45	1211	50.36	1322	17.58	1221	3.42	1221
	MIN	-45.89	1222	-79.21	1322	-17.58	1221	-3.42	1321
0.68	MAX	4.46	1222	56.01	1322	17.58	1211	7.26	1221
	MIN	-45.90	1222	-56.01	1322	-17.58	1221	-4.65	1321
0.85	MAX	4.51	1211	77.30	1322	11.10	1221	9.17	1221
	MIN	-45.91	1222	-53.73	1322	-17.58	1221	-5.88	1321

MAX/MIN FORCE VALUES FOR MEMB 409, AMONGST ALL SECT LOCATIONS
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0.51 M

MAX/MIN FORCE VALUES FOR MEMB				403. AMONGST ALL SECT LOCATIONS							
		F1	F2	DIST	NZ/	DIST	LD	FX	DIST	LD	
0.85	MAX	32.60	1013	59.11	1332	-74.24	1022	-11.11	1121	-9.62	1121
	MIN	-15.63	1211								

MIN.	-13.65	0.85	12.11	-74.66	0.00	12.12	25.98	T	0.00	12.12
	-11.11	0.65	12.11	-59.62	0.85	12.11				
404	0.00	24.75	1322	59.08	1322	0.00	1021		0.13	12.21
	MIN	-28.67	1111	-74.27	1022	0.00	1221		-0.26	11.21
	0.17	24.71	1322	54.90	1322	0.00	1022		0.13	12.21
	MIN	-28.75	1111	-69.71	1022	0.00	1211		-0.26	11.21
	0.34	24.66	1322	50.72	1322	0.00	1022		0.13	12.21
	MIN	-28.82	1111	-65.14	1022	0.00	1221		-0.26	11.21
	0.51	24.62	1322	46.35	1322	0.00	1022		0.13	12.21
	MIN	-28.57	1111	-49.35	1322	0.00	1221		-0.13	12.21
	0.68	24.57	1321	40.33	1322	0.00	1021		0.13	12.21
	MIN	-28.96	1111	-55.98	1022	0.00	1211		-0.26	11.21
	0.85	24.53	1322	38.24	1322	0.00	1022		0.13	12.21
	MIN	-29.03	1111	-51.36	1022	0.00	1211		-0.26	11.21
STAND SPACE								--	PAGE NO.	21

	MAX/MIN FORCE VALUES FOR MEMB				404, AMONGST ALL SECT LOCATIONS			
	F _x /F _y	DIST	LD		NY/NZ	DIST	LD	
								FX DIST LD
MAX.	24.75	0.00	1022		59.08	0.00	1321	
	0.00	0.00	1322		0.13	0.00	1321	15.05 C 0.00 1322
MIN.	-29.03	0.85	1311		-74.27	0.00	1032	
	0.00	0.85	1211		-0.26	0.00	1321	3.82 N 0.85 1322

[illegible]

	MAX/MIN FORCE VALUES FOR MEMB						403, AMONGST ALL SECT LOCATIONS					
	FZ	FY	DIST	LD	DIST	LD	MZ	DIST	LD	DIST	LD	
MAX.	27.14	0.00	0.00	1122	38.24	0.00	1322	0.00	1221	17.02	C	
	0.00	0.00	1022	0.14	0.00	1221	0.14	0.00	1022	0.00	1011	
MIN.	-26.95	1.63	1022	-51.36	0.00	1022	-51.36	0.00	1022	2.62	C	
	0.00	1.63	1211	-0.27	0.00	1121	-0.27	0.00	1121	1.63	1221	

	MAX	26.96	1122	3.97	1211	0.00	1022	0.14	1221
0.00	MIN	-26.53	1222	-11.85	1013	0.00	1211	-0.27	1121
0.33	MAX	26.84	1212	12.17	1121	0.00	1022	0.14	1221
	MIN	-26.62	1222	-17.03	1122	0.00	1211	-0.27	1121
0.65	MAX	26.72	1122	20.40	1211	0.00	1022	0.14	1221
	MIN	-26.71	1222	-25.77	1122	0.00	1211	-0.27	1121
0.98	MAX	26.59	1122	28.66	1211	0.00	1022	0.14	1221
	MIN	-26.87	1222	-36.97	1122	0.00	1211	-0.27	1121
1.31	MAX	26.80	1122	43.16	1121	0.00	1022	0.14	1221
	MIN	-26.80	1222	-43.16	1122	0.00	1211	-0.27	1121
1.63	MAX	26.86	1122	45.26	1211	0.00	1022	0.14	1221
	MIN	-27.00	1222	-51.74	1121	0.00	1211	-0.27	1121

	Electrical Store-AML				Electrical Store-AML			
	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FY/ FZ	DIST LD	MZ/ MY	DIST LD
MAX.	4.91	0.85 1211	79.01	0.00 1022	1.77	0.00 1221	5.85	0.00 1011
	11.10	0.00 1121	9.17	0.85 1121	-1.47	5.00 1013	4.95	4.00 1221
MIN.	-15.91	0.85 1122	-65.17	0.00 1322	-2.34	5.00 1021	-5.86	5.00 1221
	-7.29	0.85 1221	-5.88	0.85 1221				

	Electrical Store-AML				Electrical Store-AML			
	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FY/ FZ	DIST LD	MZ/ MY	DIST LD
MAX.	4.91	0.85 1211	79.01	0.00 1022	1.77	0.00 1221	5.85	0.00 1011
	11.10	0.00 1121	9.17	0.85 1121	-1.47	5.00 1013	4.95	4.00 1221
MIN.	-15.91	0.85 1122	-65.17	0.00 1322	-2.34	5.00 1021	-5.86	5.00 1221
	-7.29	0.85 1221	-5.88	0.85 1221				

1003	0.00	MAX	14.47	1013	12.20	1121	2.34	1121	4.91	1221
		MIN	-0.86	1214	-2.76	1221	-1.77	1221	-5.86	1121
	1.00	MAX	6.77	1013	3.49	1321	2.34	1121	3.14	1221
		MIN	-0.25	1214	-4.92	1021	-1.77	1221	-3.52	1121
	2.00	MAX	2.25	1321	0.74	1212	2.34	1121	1.41	1021
		MIN	-4.63	1013	-4.63	1013	-1.77	1221	-1.22	1321
	3.00	MAX	0.96	1214	1.01	1021	2.34	1121	1.16	1121
		MIN	-8.62	1013	-1.01	1321	-1.77	1221	-0.39	1221
	4.00	MAX	1.56	1214	12.62	1013	2.34	1121	5.30	1121
		MIN	-16.37	1013	-1.57	1214	-2.34	1121	-5.86	1121
	5.00	MAX	2.47	1013	35.40	1214	2.34	1121	5.86	1121
		MIN	-24.02	1013	-3.44	1214	-1.77	1221	-3.93	1221

410	0.00	MAX	13.24	1322	53.72	1322	12.43	1221	52.12	1121
		MIN	-18.89	1022	-77.32	1022	-12.79	1121	-50.66	1221
	0.82	MAX	13.22	1322	42.94	1322	12.43	1221	41.70	1121
		MIN	-18.92	1022	-65.79	1112	-12.79	1121	-40.53	1221
	1.63	MAX	13.19	1322	32.17	1322	12.43	1221	31.27	1121
		MIN	-18.95	1022	-52.45	1112	-12.79	1121	-50.40	1221
	2.45	MAX	13.17	1322	21.43	1322	12.43	1221	20.85	1121
		MIN	-20.18	1112	-37.03	1112	-12.79	1121	-20.26	1221
	3.26	MAX	13.14	1322	10.42	1322	12.43	1221	10.42	1121
		MIN	-13.74	1112	-10.52	1112	-12.79	1121	-10.52	1221
	4.08	MAX	13.12	1322	12.43	1322	12.43	1221	12.43	1121
		MIN	-23.25	1112	0.00	1112	-12.79	1121	0.00	1121

MAX/MIN FORCE VALUES FOR MEMB 1003, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FY/ FZ	DIST LD	MZ/ MY	DIST LD	
MAX.	14.47	0.00 1013	32.80	5.00 1013					
	2.34	0.00 1121	5.84	5.00 1121					
MIN.	-24.02	5.00 1013	-4.92	1.00 1021					
	-1.77	5.00 1221	-5.86	0.00 1121					

MAX/MIN FORCE VALUES FOR MEMB 410, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FY/ FZ	DIST LD	MZ/ MY	DIST LD	
MAX.	13.24	0.00 1322	53.72	0.00 1322					
	12.43	0.00 1221	52.12	0.00 1121					
MIN.	-23.25	4.08 1112	-77.32	0.00 1022					
	-12.79	4.08 1121	-50.66	0.00 1221					

1004	0.00	MAX	23.10	1013	34.65	1013	0.00	1321	0.00	1021
		MIN	-1.81	1214	-2.72	1214	0.00	1021	0.00	1321
	0.60	MAX	18.48	1013	22.18	1013	0.00	1321	0.00	1021
		MIN	-1.45	1214	-1.74	1214	0.00	1021	0.00	1321
	1.20	MAX	13.86	1013	12.48	1013	0.00	1321	0.00	1021
		MIN	-0.98	1214	-0.98	1214	0.00	1021	0.00	1321
	1.80	MAX	9.24	1013	5.54	1013	0.00	1321	0.00	1021
		MIN	-0.73	1214	-0.44	1214	0.00	1021	0.00	1321
	2.40	MAX	4.62	1013	1.39	1013	0.00	1321	0.00	1021
		MIN	-0.36	1214	-0.11	1214	0.00	1021	0.00	1321
	3.00	MAX	0.00	1021	0.00	1022	0.00	1321	0.00	1021
		MIN	0.00	1321	0.00	1322	0.00	1021	0.00	1321

1001	0.00	MAX	0.00	1221	0.00	1222	0.00	1121	0.00	1022
		MIN	0.00	1121	0.00	1112	0.00	1221	0.00	1212
	0.60	MAX	0.36	1214	1.39	1013	0.00	1121	0.00	1022
		MIN	-4.62	1013	-0.11	1214	0.00	1221	0.00	1211
	1.20	MAX	0.73	1214	5.54	1013	0.00	1121	0.00	1121
		MIN	-9.24	1013	-0.44	1214	0.00	1221	0.00	1221
	1.80	MAX	1.09	1214	12.48	1013	0.00	1121	0.00	1121
		MIN	-13.86	1013	-0.98	1214	0.00	1221	0.00	1221
	2.40	MAX	1.45	1214	22.18	1013	0.00	1121	0.00	1121
		MIN	-18.48	1013	-1.74	1214	0.00	1221	0.00	1221
	3.00	MAX	1.81	1214	34.65	1013	0.00	1121	0.00	1121
		MIN	-23.10	1013	-2.72	1214	0.00	1221	0.00	1221

MAX/MIN FORCE VALUES FOR MEMB 1004, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FY/ FZ	DIST LD	MZ/ MY	DIST LD	
MAX.	23.10	0.00 1013	34.65	0.00 1013					
	0.00	0.00 1321	0.00	0.00 1021					
MIN.	-1.81	0.00 1214	-2.72	0.00 1214					
	0.00	3.00 1021	0.00	0.00 1321					

MAX/MIN FORCE VALUES FOR MEMB 1001, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FY/ FZ	DIST LD	MZ/ MY	DIST LD	
MAX.	1.81	3.00 1214	34.65	3.00 1013					
	0.00	0.00 1121	0.00	3.00 1121					
MIN.	-23.10	3.00 1013	-2.72	3.00 1214					
	0.00	3.00 1221	0.00	3.00 1221					

2001	0.00	MAX	0.00	1112	0.00	1112	0.00	1121	0.00	1121
		MIN	0.00	1321	0.00	1321	0.00	1121	0.00	1121
	0.60	MAX	4.39	1214	1.19	1013	0.00	1121	0.00	1121
		MIN	-4.92	1013	-1.19	1214	0.00	1121	0.00	1121
	1.20	MAX	0.92	1214	5.54	1013	0.00	1121	0.00	1121
		MIN	-9.24	1013	-0.44	1214	0.00	1121	0.00	1121
	1.80	MAX	1.09	1214	12.48	1013	0.00	1121	0.00	1121
		MIN	-13.86	1013	-0.98	1214	0.00	1121	0.00	1121
	2.40	MAX	1.45	1214	22.18	1013	0.00	1121	0.00	1121
		MIN	-18.48	1013	-1.74	1214	0.00	1121	0.00	1121
	3.00	MAX	1.81	1214	34.65	1013	0.00	1121	0.00	1121
		MIN	-23.10	1013	-2.72	1214	0.00	1121	0.00	1121

1002	0.00	MAX	24.02	1013	32.80	1013	1.77	1321	5.85	1021
		MIN	-2.17	1214	-3.44	1214	-2.34	1021	-3.93	1321
	1.00	MAX	16.32	1013	12.62	1013	1.77	1321	3.14	1021
		MIN	-1.29	1214	-1.01	1214	-2.34	1021	-1.16	1321
	2.00	MAX	8.86	1214	1.01	1121	1.77	1321	1.16	1021
		MIN	-8.86	1214	-1.01	1221	-2.34	1021	-1.16	1321
	3.00	MAX	2.87	1121	0.74	1212	1.77	1321	0.40	1321
		MIN	-2.26	1221	-0.63	1013	-2.34	1021	-1.41	1121
	4.00	MAX	0.25	1214	3.50	1221	1.77	1321	3.14	1321
		MIN	-6.77	1013	-4.93	1121	-2.34	1021	-3.52	1021
	5.00	MAX	0.86	1214	12.21	1021	1.77	1321	4.91	1321
		MIN	-14.47	1013	-2.78	1321	-2.34	1021	-5.86	1021

MAX/MIN FORCE VALUES FOR MEMB 2001, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FY/ FZ	DIST LD	MZ/ MY	DIST LD	
MAX.	1.81	3.00 1214	34.65	3.00 1013					
	0.00	0.00 1221	0.00	3.00 1221					
MIN.	-23.10	3.00 1013	-2.72	3.00 1214					
	0.00	3.00 1121	0.00	3.00 1121					

MAX/MIN FORCE VALUES FOR MEMB 1002, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FY/ FZ	DIST LD	MZ/ MY	DIST LD	
MAX.	24.02	0.00 1013	32.80	0.00 1013					

Electrical Store ANL									
MAX/MIN FORCE VALUES FOR MEMB									
3004, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
	FZ			MY					
MAX.	0.00	0.00	1212	0.00	0.00	1212	0.00	0.00	1212
MIN.	0.00	0.00	1212	0.00	0.00	1212	0.00	0.00	1212
STAAD SPACE									
4001									
MAX.	0.00	0.00	1212	0.00	0.00	1212	0.00	0.00	1212
MIN.	0.00	0.00	1212	0.00	0.00	1212	0.00	0.00	1212
STAAD SPACE									
4002									
MAX.	0.00	0.00	1212	0.00	0.00	1212	0.00	0.00	1212
MIN.	0.00	0.00	1212	0.00	0.00	1212	0.00	0.00	1212
STAAD SPACE									
4003									
MAX.	0.00	0.00	1212	0.00	0.00	1212	0.00	0.00	1212
MIN.	0.00	0.00	1212	0.00	0.00	1212	0.00	0.00	1212
STAAD SPACE									

Electrical Store ANL									
MAX/MIN FORCE VALUES FOR MEMB									
4003, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
	FZ			MY					
MAX.	0.71	0.00	1321	4.46	5.00	1021	6.50	5.00	1021
MIN.	-1.62	5.00	1021	-5.81	0.00	1022	15.61	0.00	1214
STAAD SPACE									
4004									
MAX.	0.00	0.00	1321	0.00	0.00	1022	0.00	0.00	1322
MIN.	0.00	0.00	1321	0.00	0.00	1022	0.00	0.00	1322
STAAD SPACE									
4005									
MAX.	0.00	0.00	1321	0.00	0.00	1022	0.00	0.00	1322
MIN.	0.00	0.00	1321	0.00	0.00	1022	0.00	0.00	1322
STAAD SPACE									

Electrical Store ANL									
MAX/MIN FORCE VALUES FOR MEMB									
4004, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
	FZ			MY					
MAX.	0.00	0.00	1111	0.00	2.40	1212	0.00	0.00	1221
MIN.	-15.05	0.00	1112	0.00	0.00	1322	0.36	0.00	1114
STAAD SPACE									
8001									
MAX.	0.00	0.00	1111	0.00	0.00	1322	0.00	0.00	1322
MIN.	0.00	0.00	1111	0.00	0.00	1322	0.00	0.00	1322
STAAD SPACE									
8002									
MAX.	0.00	0.00	1111	0.00	0.00	1322	0.00	0.00	1322
MIN.	0.00	0.00	1111	0.00	0.00	1322	0.00	0.00	1322
STAAD SPACE									

Electrical Store ANL									
MAX/MIN FORCE VALUES FOR MEMB									
8001, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
	FZ			MY					
MAX.	0.40	0.00	1001	0.00	0.00	1001	5.28	0.00	1322
MIN.	-0.40	0.00	1001	0.00	0.00	1322	23.52	0.00	1012
STAAD SPACE									
8002									
MAX.	0.00	0.00	1114	0.00	0.00	1322	0.00	0.00	1322
MIN.	0.00	0.00	1114	0.00	0.00	1322	0.00	0.00	1322
STAAD SPACE									

Electrical Store ANL									
MAX/MIN FORCE VALUES FOR MEMB									
4002, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
	FZ			MY					
MAX.	1.61	0.00	1321	4.46	0.00	1321	6.48	0.00	1121
MIN.	-0.69	5.00	1021	-5.81	5.00	1022	15.61	5.00	1214
STAAD SPACE									
4003									
MAX.	0.00	0.00	1321	0.00	0.00	1321	0.00	0.00	1321
MIN.	0.00	0.00	1321	0.00	0.00	1321	0.00	0.00	1321
STAAD SPACE									

Electrical Store ANL									
MAX/MIN FORCE VALUES FOR MEMB									
8002, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
	FZ			MY					
MAX.	0.00	0.00	1114	0.00	0.00	1322	0.00	0.00	1322
MIN.	0.00	0.00	1114	0.00	0.00	1322	0.00	0.00	1322
STAAD SPACE									

Electrical Store-ANL

MAX.	0.40	0.00	1001	0.00	0.00	1001	7.84	T	0.00	1221
MIN.	0.00	0.00	1001	0.00	6.98	1322	19.13	T	6.98	1013
	-0.40	6.98	1114	0.00	6.98	1322				
	0.00	0.00	6.98	1322						

8003	0.00	MAX	0.40	1114	0.00	1322	0.00	1322	0.00	1322
	MIN	0.26	1322	0.00	1322	0.00	1322	0.00	1322	
	1.40	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322	
	2.79	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322	
	4.19	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322	
	5.58	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322	
	6.98	MAX	-0.40	1114	0.00	1322	0.00	1322	0.00	1322
	MIN	-0.40	1114	0.00	1322	0.00	1322	0.00	1322	

MAX/MIN FORCE VALUES FOR MEMB 8003, AMONGST ALL SECT LOCATIONS

	FZ	FY	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.40	0.00	1001	0.00	0.00	1001	6.40	T	6.98	1321
MIN.	-0.40	6.98	1114	0.00	6.98	1322	19.13	T	0.00	1013
	0.00	0.00	6.98	1322						

8004	0.00	MAX	0.40	1114	0.00	1322	0.00	1322	0.00	1322
	MIN	0.26	1322	0.00	1322	0.00	1322	0.00	1322	
	1.40	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322	
	2.79	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322	
	4.19	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322	
	5.58	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322	
	6.98	MAX	-0.40	1114	0.00	1322	0.00	1322	0.00	1322
	MIN	-0.40	1114	0.00	1322	0.00	1322	0.00	1322	

STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB 8004, AMONGST ALL SECT LOCATIONS

	FZ	FY	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.40	0.00	1001	0.00	0.00	1001	5.28	T	6.98	1322
MIN.	-0.40	6.98	1114	0.00	6.98	1322	23.52	T	0.00	1012
	0.00	0.00	6.98	1322						

***** END OF FORCE ENVELOPE FROM INTERNAL STORAGE *****

176. LOAD LIST 2001 2002 2011 TO 2014

177. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACE ALL

STAAD SPACE

--- PAGE NO. 35

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
102	2001	0.0188	-0.8712	-0.3274	0.0008	-0.0001	0.0030
	2002	0.0188	-0.8712	-0.3274	0.0007	-0.0001	0.0031
	2011	0.0188	-0.8712	-0.3274	0.0007	-0.0001	0.0031
	2012	0.0188	-0.8712	-0.3274	0.0007	-0.0001	0.0031
	2013	0.0188	-0.8712	-0.3274	0.0007	-0.0001	0.0031
	2014	0.0188	-0.8712	-0.3274	0.0007	-0.0001	0.0031
104	2001	0.0117	-0.6250	-0.1781	0.0004	-0.0001	0.0025
	2002	0.0117	-0.6250	-0.1781	0.0004	-0.0001	0.0025
	2011	0.0117	-0.6250	-0.1781	0.0004	-0.0001	0.0025
	2012	0.0117	-0.6250	-0.1781	0.0004	-0.0001	0.0025
	2013	0.0117	-0.6250	-0.1781	0.0004	-0.0001	0.0025
	2014	0.0117	-0.6250	-0.1781	0.0004	-0.0001	0.0025

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2002	0.0208	-0.7828	0.3348	0.0014	0.0013	0.0024
2011	0.0223	-0.7747	8.2541	0.0045	0.0012	0.0019
2012	0.0184	-0.7744	8.2544	0.0051	0.0011	0.0014
2013	0.0319	-1.2202	0.4134	0.0020	0.0017	0.0038
2014	0.0121	-0.3861	0.2817	0.0007	0.0011	0.0011
110	2001	0.0188	0.2761	-0.0009	0.0001	0.0030
	2002	0.0189	0.2582	-0.0008	0.0001	0.0031
	2011	0.0190	6.7860	0.0107	0.0001	0.0031
	2012	0.0192	6.8651	0.0104	-0.0001	0.0031
	2013	0.0117	0.1268	-0.0011	0.0002	0.0027
	2014	0.0117	0.1268	-0.0011	0.0002	0.0027
502	2001	-0.0188	-0.3274	0.0008	0.0001	-0.0030
	2002	-0.0189	-0.3274	0.0007	0.0001	-0.0031
	2011	-0.0186	-0.3274	0.0008	0.0001	-0.0031
	2012	-0.0186	-0.3274	0.0008	0.0001	-0.0031
	2013	-0.0149	-0.3274	0.0011	0.0002	-0.0027
	2014	-0.0117	-0.3274	0.0004	0.0001	-0.0025
504	2001	-0.0245	-0.9196	0.0016	-0.0014	-0.0028
	2002	-0.0208	-0.7828	0.0014	-0.0013	-0.0024
	2011	-0.0223	-0.7747	0.0045	-0.0012	-0.0019
	2012	-0.0184	-0.7744	0.0051	-0.0011	-0.0014
	2013	-0.0319	-1.2202	0.0020	-0.0017	-0.0038
	2014	-0.0121	-0.3861	0.0007	-0.0011	-0.0011
108	2001	0.0242	-0.9119	0.0016	-0.0014	0.0028
	2002	0.0206	-0.7790	0.0014	-0.0013	0.0024
	2011	0.0165	-0.7761	0.0039	-0.0013	0.0017
	2012	0.0117	-0.6715	0.0035	-0.0010	0.0011
	2013	0.0317	-1.2125	-0.4551	-0.0016	0.0038
	2014	0.0118	-0.3784	-0.3274	-0.0011	-0.0011
508	2001	-0.0118	-0.3784	-0.3274	-0.0011	-0.0011
	2002	-0.0118	-0.3784	-0.3274	-0.0011	-0.0011
	2011	-0.0165	-0.7761	0.0039	-0.0013	0.0017
	2012	-0.0117	-0.6715	0.0035	-0.0010	0.0011
	2013	-0.0317	-1.2125	-0.4551	-0.0016	0.0038
	2014	-0.0118	-0.3784	-0.3274	-0.0011	-0.0011
510	2001	-0.0118	-0.3784	-0.3274	-0.0011	-0.0011
	2002	-0.0118	-0.3784	-0.3274	-0.0011	-0.0011
	2011	-0.0189	-0.8156	0.2582	-0.0001	-0.0031
	2012	-0.0190	-1.5144	6.7860	0.0107	-0.0001
	2013	-0.0192	-1.5289	6.8651	0.0104	0.0001
	2014	-0.0149	-0.7251	0.3766	-0.0011	-0.0002
	2015	-0.0117	-0.6200	0.1268	-0.0005	-0.0025

STAAD SPACE

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PAGE NO.

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
201	2001	0.0000	0.0000	0.0000	-0.0014	-0.0003	-0.0005
	2002	0.0000	0.0000	0.0000	-0.0012	-0.0003	-0.0005
	2011	0.0000	0.0000	0.0000	-0.0014	-0.0003	-0.0005
	2012	0.0000	0.0000	0.0000	-0.0012	-0.0003	-0.0005
	2013	0.0000	0.0000	0.0000	-0.0018	-0.0001	0.0021
202	2014	0.0000	0.0000	0.0000	-0.0008	0.0003	0.0045
	2001	0.0006	-0.0558	-0.2843	0.0009	-0.0001	0.0010
	2002	0.0006	-0.0488	-0.2394	0.0007	-0.0001	0.0010
	2011	0.0006	-0.0488	-0.2394	0.0007	-0.0001	0.0010
	2012	0.0006	-0.0488	-0.2394	0.0007	-0.0001	0.0010
203	2013	0.0005	-0.0693	6.7390	0.0109	-0.0005	0.0010
	2014	0.0005	-0.0694	6.7000	0.0114	-0.0002	0.0010
	2001	-0.0007	-0.0677	-0.3562	0.0011	-0.0002	0.0007
	2002	-0.0045	-0.0764	-0.1506	0.0004	-0.0001	0.0004
	2011	-0.0405	-0.0668	-0.1458	0.0016	0.0018	0.0005
204	2012	-0.0443	-0.0668	-0.1458	0.0014	0.0020	0.0006
	2013	-0.0447	-0.0677	-0.1458	0.0014	0.0020	0.0006
	2014	-0.0487	-0.0585	-0.1458	0.0014	0.0020	0.0006
	2001	-0.0329	-0.0941	-0.2202	0.0020	0.0017	0.0005
	2002	-0.0349	-0.0428	-0.0972	0.0008	0.0025	0.0011
205	2011	0.0053	-0.1770	-0.0615	0.0016	0.0014	0.0012
	2012	0.0044	-0.1544	-0.0444	0.0014	0.0013	0.0010
	2013	0.0091	0.2315	7.9229	0.0045	0.0013	0.0009
	2014	0.0087	0.1930	7.9672	0.0050	0.0011	0.0008
	2001	0.0063	-0.2831	-0.0711	0.0007	0.0011	0.0006
206	2012	0.0081	-0.0841	-0.0071	0.0007	0.0011	0.0006
	2001	0.0781	-0.3422	-0.0141	0.0010	0.0001	-0.0022
	2002	0.0781	-0.3426	-0.0141	0.0010	0.0001	-0.0022
	2011	0.0768	-0.0066	8.0056	0.0019	0.0001	-0.0019
	2012	0.0760	-0.0808	8.0630	0.0023	0.0001	-0.0020
207	2013	0.0922	-0.3510	0.0030	0.0013	0.0001	-0.0024
	2014	0.0773	-0.1375	-0.0256	0.0005	0.0001	-0.0022
	2001	0.1084	-0.3595	-0.0257	0.0000	0.0000	-0.0029
	2002	0.1036	-0.3146	-0.0128	0.0000	0.0000	-0.0028
	2011	0.1035	-0.0816	8.0055	-0.0001	0.0000	-0.0027
208	2012	0.0966	-0.1840	8.0649	-0.0001	0.0000	-0.0026

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

	PASS	85-4,6 (T)	2013
8002 ST UA90x90x6	15,94 T	0,00	0,00
8003 ST UA90x90x6	PASS	85-4,6 (T)	0,04
	15,94 T	0,00	0,00
8004 ST UA90x90x6	PASS	85-4,6 (T)	0,054
	19,60 T	0,00	0,00

END OF TABULATED RESULT OF DESIGN

182. STEEL TAKE OFF ALL
STAAD SPACE

STEEL TAKE OFF

STEEL TAKE-OFF

PROFILE	LENGTH(METE)	WEIGHT(KN)
ISECT: TAPERED	201	25.779
ISECT: TAPERED	3001	25.081
ST UA90X90X6	27.92	2.274
	TOTAL =	53.134

END OF DATA FROM INTERNAL STORAGE

183. FINISH

```
*****
END OF THE STAAD.PRO RUN
*****
```

*** DATE= APR 28.2007 TIME= 15:20:42 ***

2 FOUNDATION DESIGN CHECK

Document No.: **DM001-E-AAO-CDP-DR-DCC-373904**

Rev A3

Date: 02 May 2007

Page

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Auxiliary Depot Jebel - Building 11	Date	Apr '07	Job No.	068010
	Footing Summary				

Designed by Rudy Sheet No.

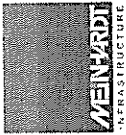
Soil Bearing Pressure	=	450	kN/m ²
Soil Density	=	20	kN/m ³
Concrete Density	=	24	kN/m ³
Footing Depth below slab soffit	=	1	m
Column Stump Area	=	0.25	m ²

Individual Footing

Column	Column Reaction (ULS)	Soil Load (kN)	Footing Load (kN)	Total SLS Load (kN)	ULS Load (kN)	Footing size			Area of Footing (m ²)	Soil Pressure (kN/m ²)	Factor	Remarks
						L (m)	B (m)	H (m)				
201	85	34.2	28.224	133	172	1.400	1.400	0.600	1.96	68	6.62	OK

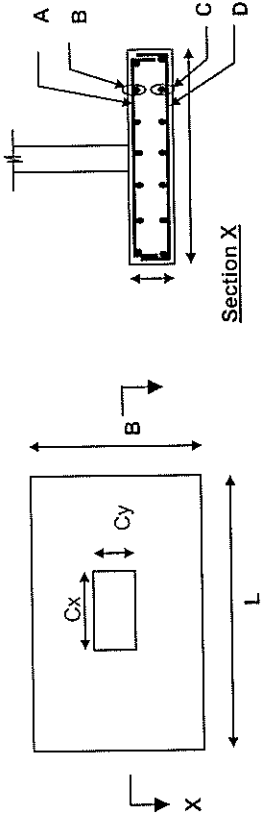
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Auxiliary Depot Jebel - Building 11 Footing Detail	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Summary



Footing Type	L	B	H	Soil Bearing pressure (kPa)	A	B	C	D	Spacing (mm)
F3	1400	1400	600	450	10T12	10T12	10T16	10T16	143

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Auxiliary Depot Jebel - Building 11 Footing F3 For soil bearing pressure = 150kPa	Date	Apr '07	Job No.	068010.01
		Designed by	Rudy	Sheet No.	

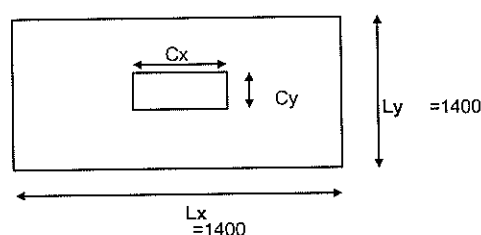
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	35	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	1400	mm
Length of Footing	L_x	=	1400	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	16	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_c	=	12	mm
Comp. bar diameter (y-dir)	D_c	=	12	mm
Effect depth (x-dir)	d_x	=	542	mm
Effect depth (y-dir)	d_y	=	526	mm
Area of footing	A	=	1.96	m ²



3. Loadings

Safe bearing pressure	p_b	=	450	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	882.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	p_{max}	=	435.60	KN/m
Net pressure (ULS)	p_{max}	=	653.4	KN/m
Shear force (x-dir)	V_x	=	434.51	kN
Shear force (y-dir)	V_y	=	434.51	kN
Moment (x-dir)	M_x	=	103.20	kNm
Moment (y-dir)	M_y	=	103.20	kNm
Steel required (x-dir)	A_{sreq}	=	1092	mm ²
Steel provided (x-dir)	A_{sprov}	=	2011	mm ²
Steel required (y-dir)	A_{sreq}	=	1092	mm ²
Steel provided (y-dir)	A_{sprov}	=	2011	mm ²
Top bars required (x-dir)	A'_{sreq}	=	1092	mm ²
Top bars provided (x-dir)	A'_{sprov}	=	1131	mm ²
Top bars required (y-dir)	A'_{sreq}	=	1092	mm ²
Top bars provided (y-dir)	A'_{sprov}	=	1131	mm ²

Nos	ϕ	Spacing	
10	16	143	x-dir
10	16	143	y-dir
10	12	143	x-dir
10	12	143	y-dir

Calculation/Sketch

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Site	Auxiliary Depot Jebel - Building 11	Date	Apr '07	Job No.	068010.01
	Footing F3	Designed by	Rudy	Sheet No.	
	For soil bearing pressure = 150kPa				

5. Shear Check

Checking for punching shear:

Shear force at face V_f = 1186.31 kN
Shear stress at face v_f = 1.22 N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) V_c = 448.88 kN
Shear stress at face (x-dir) v_c = 0.59 N/mm² O.K.

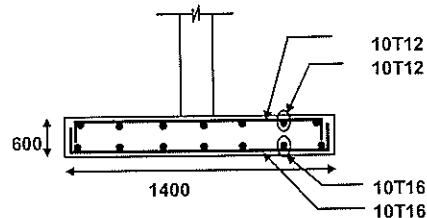
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) V_c = 448.88 kN
Shear stress at face (y-dir) v_c = 0.61 N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	526	0.00	0.00	0.46	0.92	No Shear Links
2d	1052	0.00	0.00	0.46	0.46	No Shear Links

6. Detailing



3 CONNECTIONS DESIGN

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Rev A3

Date: 02 May 2007

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Calculation/Sketch

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Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Column Splice Connection Design

From the STAAD Pro analysis

Fxc =	47	kN (compression)
Fxt =	22	kN (tension)
Fy =	20	kN
Mz =	75	kNm

Total Tension

$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$
where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$= 0.5a \cdot p_c \cdot b$
where b = stiff bearing length
= web thickness + 2 * plate thickness

Moment Capacity

$M_c = [n \cdot T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$

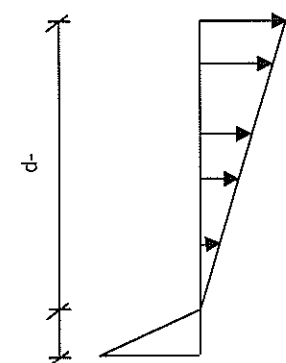
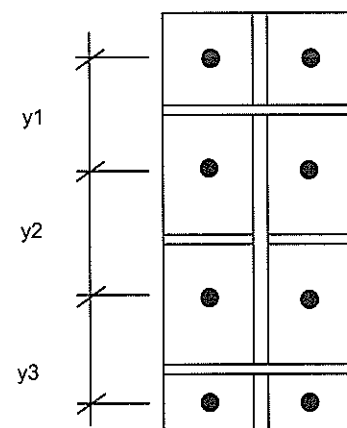
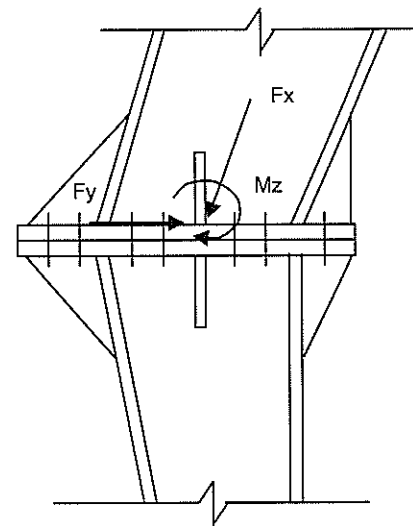
For Gr 8.8 bolt

Bolt size =	20 mm
T_b =	110 kN
V_b =	91.9 kN
n =	2
n total =	12
web, t_w =	8 mm
plate, t_p =	10 mm
b =	28 mm

y1	127	Trial a	=	84 mm
y2	147	T	=	381.6 kN
y3	127	C	=	381.6 kN
y4	0	T-C	=	0.00 kN
y5	0	M_c	=	95.8 kNm
y6	0			
Depth, d	400			

Tension force on the bolt, T_1 = $(M_z / M_c) \cdot T_b$
= 86 kN

$[T_1 + (F_{xt} / n_{total})] / T_b + F_y / (n_{total} \cdot V_b) = 0.82$
< 1.4 Ok



$p_c = 325 \text{ N/mm}^2$

Calculation/Sketch

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Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s =	6 mm	(up to 15mm thick)
Weld strength, pw =	1.05 kN/mm	(based on longitudinal capacity)
Weld length required =	$(T1 + 0.5 \cdot Fxt) / pw$	
=	92 mm	
=	46 mm per face	

Calculation/Sketch

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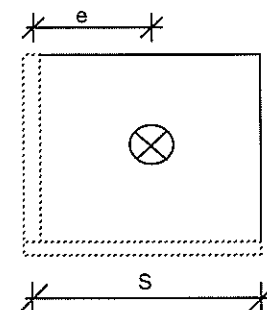


Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

From the plate analysis (refer attached) using

Prying Force, $T_o = 10 \text{ kN}$

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75	0.020
100	0.027
125	0.035
150	0.040



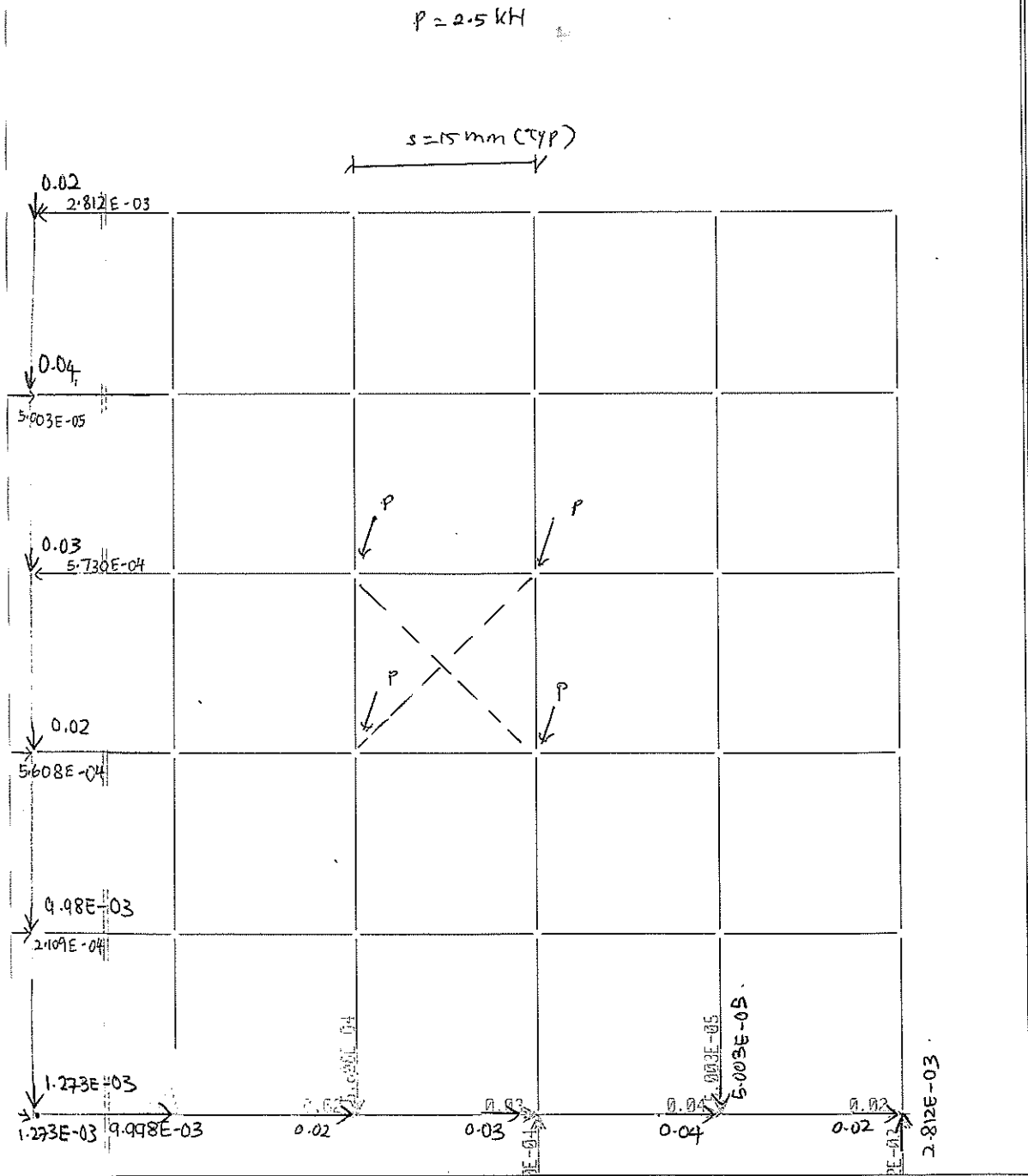
Actual - Prying force, T	=	79 kN
Side, S	=	75 mm
Bending Moment, M	=	$(T/T_o) * M_o$
	=	0.160 kNm

Bolt Diameter, ϕ	=	20 mm
Bolt distance, e	=	36 mm
Thickness of plate, t_p	=	10 mm
p_y allow	=	355 N/mm ²

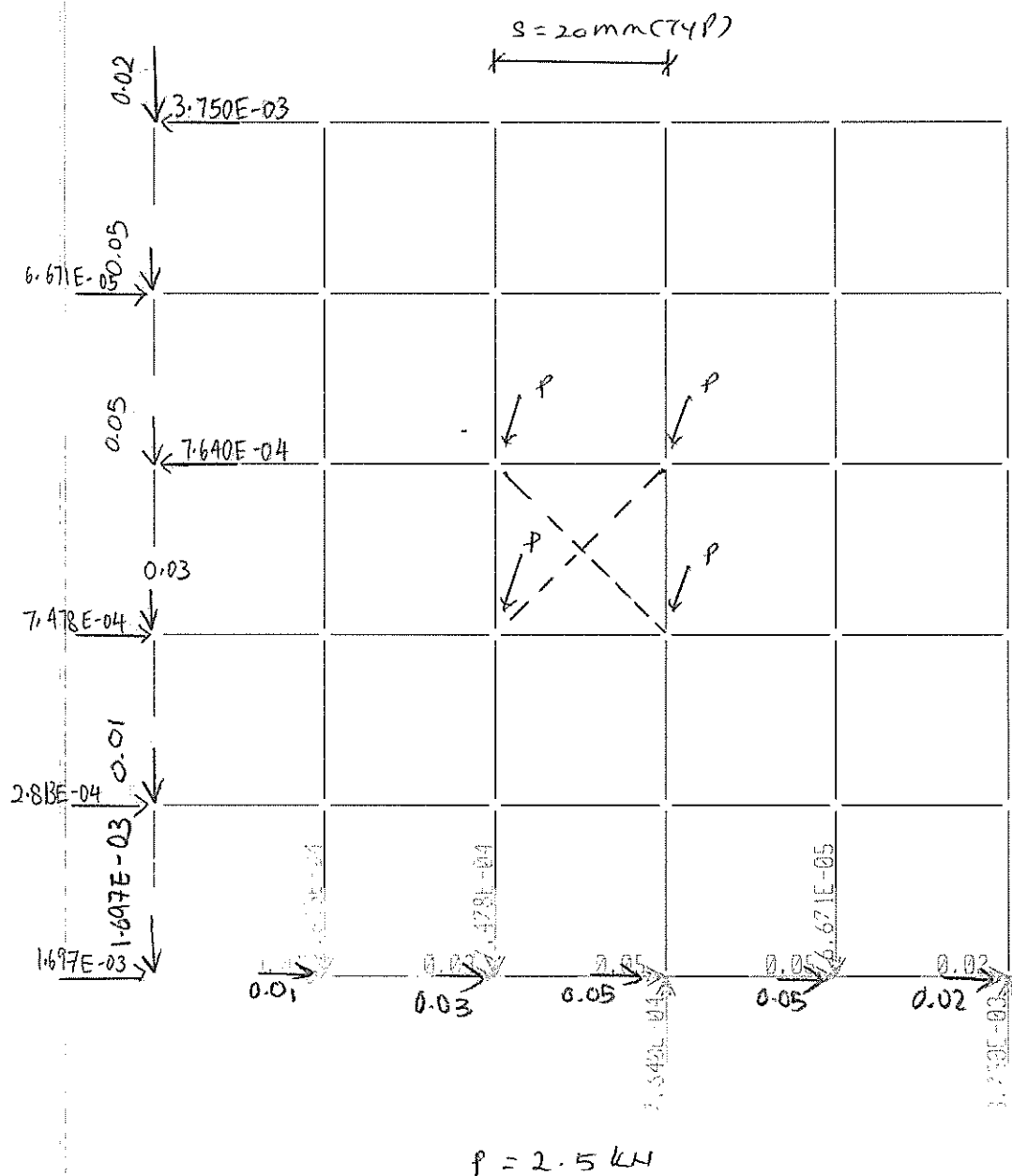
Effective width of the plate resisting the bending moment, b_p	=	$\min(2e + \phi, S)$
	=	75 mm

Section of modulus of the plate, Z_p	=	$b_p * t_p^2 / 6$
	=	1250 mm ²

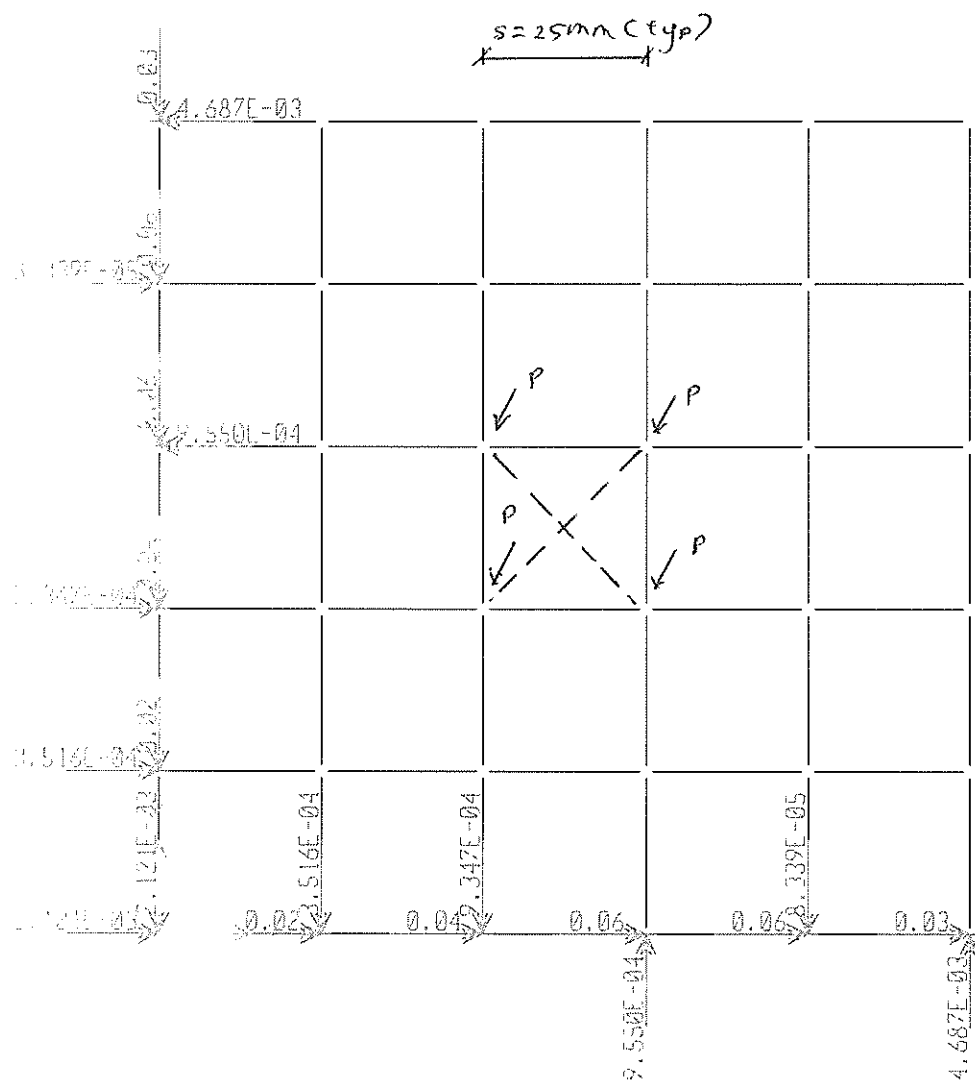
$M / Z_p = 128 \text{ N/mm}^2 \quad \text{Ok}$



Bolt Prying Action on Splice Plate
(Plate Width = 75 mm)

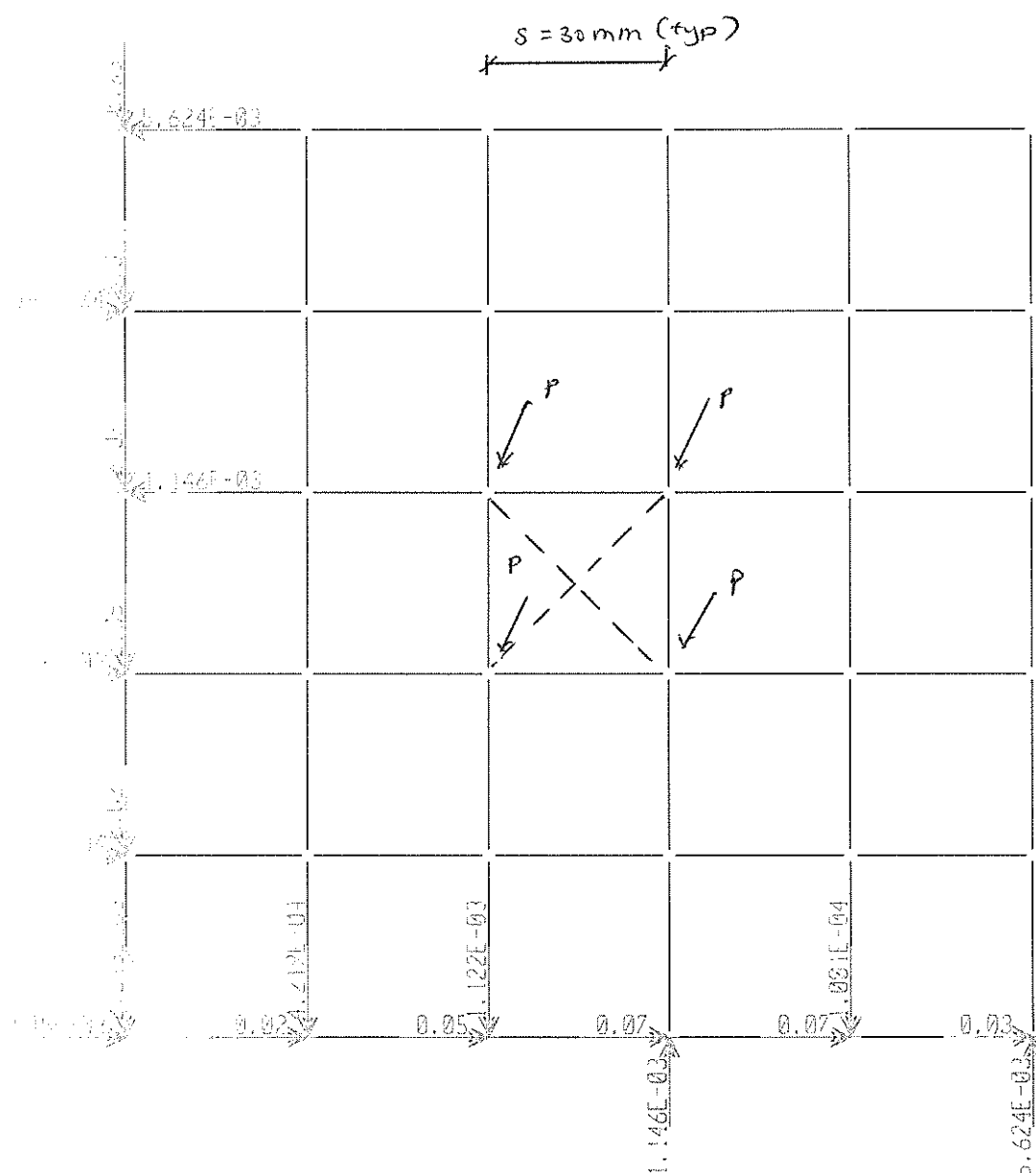


BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 100mm)



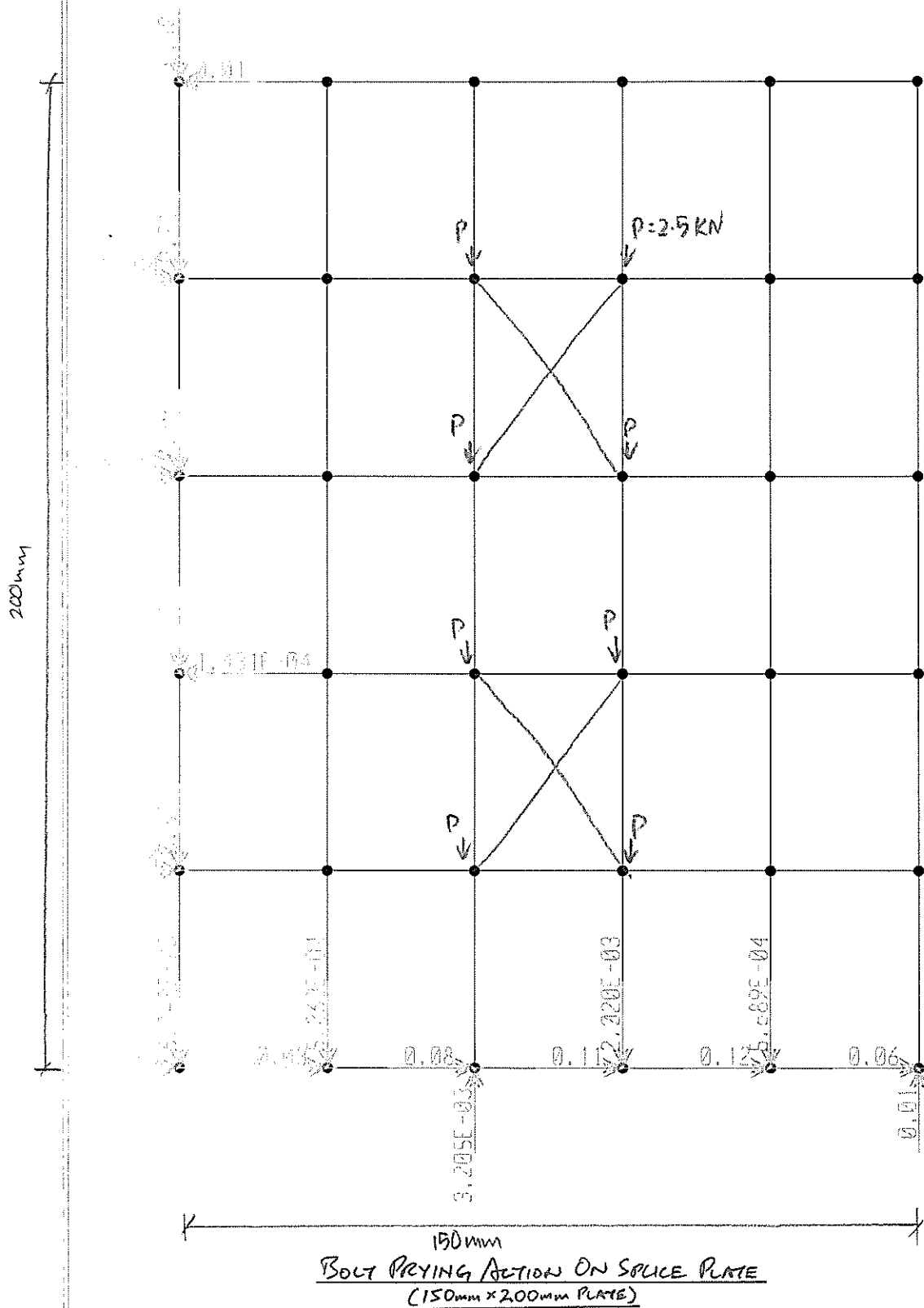
$$P = 2.5 \text{ kN}$$

BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 125mm)



$$P = 2.5 \text{ kN}$$

BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 150mm)



Calculation/Sketch

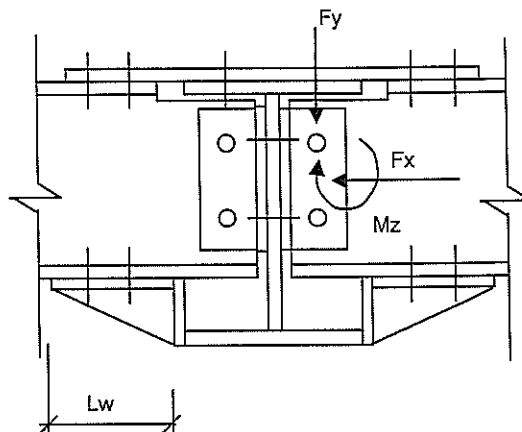
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Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Secondary Beam Connection Design

Roof SDL	=	0.8 kN/m ²
Self weight	=	0.5 kN/m ²
WL	=	0.6 kN/m ² (downward)
WL	=	0.6 kN/m ² (downward)
	=	1.4 kN/m ² (upward)
wuls	-	1.4*DL+1.6LL = 2.78 kN/m ²
	-	1.2*(DL+LL+WL) = 3.00 kN/m ²
	-	1.4WL - DL = 0.66 kN/m ²
loading width, s	=	3.8 m



For internal span (simply supported span)

- top and bottom plate not required

max span, L = 8 m

Axial load for earthquake, Pa = (11.5+11.5)*(7.1) / 12 rows of secondary beams / tie beams
= 13.6 kN

Vmax = 0.5*wuls*L*s
Downward forces = 45.6 kN
Upward forces = 10.0 kN

Bolt diameter, ϕ = 16 mm
Shear capacity, Vb = 58.9 kN
Bolt tensile capacity, Tb = 70.7 kN

Check shear on bolt connected to secondary beam's web

No of bolt, n = 2
Vmax / n = 22.8 kN
Pa / n = 6.8 kN
Resultant shear, V = 23.8 kN Ok

Check tensile capacity of bolt connected to main beam's web

Eccentricity of load, e = 60 mm
Moment due to eccentricity, Me = Vmax*e
= 2.736 kNm
Spacing between bolt, db = 150 mm
Tensile force on the bolt, T = Me/db
= 18.2 kN Ok

Calculation/Sketch

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Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
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For the cantilever end

- Assume shear due to bending is resisted by top and bottom bolt
- Assume vertical and horizontal shear is resisted by bolt on the web

Span length, L	=	4 m
Mmax	=	$0.5 \cdot wuls \cdot L^2 \cdot s$
Downward	=	91.2 kNm
Upward	=	20.1 kNm

Vmax	=	$wuls \cdot L \cdot s$
Downward	=	45.6 kN
Upward	=	10.0 kN

Check bolts on the flanges

Beam depth - flange thickness, Df	=	292 mm
Mmax bending moment (incl. moment due to ecc), Me	=	93.9 kNm

Shear on the flange, Vf	=	Me / Df
	=	321.7 kN

No of bolt provided	=	4
Shear per bolt	=	80.4 kN
Bolt diameter, ϕ	=	20 mm
Shear capacity, Vb	=	91.9 kN

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	22.8 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, Vb	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	150 mm
Tensile load per bolt	=	18.2 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, Vb	=	70.7 kN